American Terawatt

HVDC DESIGN

HVDC Design

Design notes for VSC HVDC transmission

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INTERNAL

Revision history

REV.	DATE	AUTHOR	DESCRIPTION
0.1	Oct. 19, 2026	Eugen Starschich	Initial document: creepage distance of DC OHL insulators
0.2	Oct. 20, 2026	Eugen Starschich	Added AC power transmission: line parameters, two-port and distributed representation
0.3	Oct. 20, 2026	Eugen Starschich	Split the AC material into the general representation, overhead lines, cables and the worked example
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1.0	Oct. 21, 2026	Eugen Starschich	Added DC fault ride-through: recovery sequence, records and design alternatives of the Pugalur–Thrissur link

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1 Abbreviations

Abbreviation	Meaning
AAAC	All-aluminium alloy conductor
AAC	All-aluminium conductor
AC	Alternating current
ACSR	Aluminium conductor steel reinforced
ACSS	Aluminium conductor steel supported
ASTM	American Society for Testing and Materials
AVR	Automatic voltage regulator
CB	Circuit breaker
CIGRE	International Council on Large Electric Systems
DC	Direct current
DS	Disconnector (disconnect switch)
ESDD	Equivalent salt deposit density
FACTS	Flexible AC transmission system
GMD	Geometric mean distance
GMR	Geometric mean radius
HVDC	High-voltage direct current
IEC	International Electrotechnical Commission
IGBT	Insulated-gate bipolar transistor
LCC	Line-commutated converter
MMC	Modular multilevel converter
MOV	Metal-oxide varistor
NSDD	Non-soluble deposit density
OHL	Overhead line
PSS	Power system stabiliser
SCL	Short-circuit level
SCR	Short-circuit ratio
SIL	Surge-impedance loading
SPS	Site pollution severity
SSCI	Sub-synchronous control interaction
SSR	Sub-synchronous resonance
STATCOM	Static synchronous compensator
TB	Technical brochure
TS	Technical specification
USCD	Unified specific creepage distance
VSC	Voltage-sourced converter
XLPE	Cross-linked polyethylene

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2 Symbols

Quantities per unit length carry a prime, R', L', G', C'; the same letter without the prime is the quantity of the whole line. Complex quantities are underlined. Terminal quantities take the subscript S at the sending end and R at the receiving end, and voltages are line-to-line unless stated otherwise.

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Symbol	Description
$\underline{A}, \underline{B}, \underline{C}, \underline{D}$	Two-port (ABCD) parameters of the line
C'	Capacitance per unit length
d	Conductor diameter
D	Phase spacing of the tower head
f	System frequency; f_{er} the electrical resonance frequency
G'	Conductance per unit length
GMD	Geometric mean distance between the phases, D_m
GMR	Geometric mean radius of a sub-conductor; GMR_b of the bundle
H	Inertia constant of a synchronous machine
I	Phase current; $\underline{I}_S$, $\underline{I}_R$ at the terminals
k	Degree of series compensation, $X_C = k \cdot X_L$
K_s	Synchronising power coefficient, $\partial P / \partial \delta$
l	Line length; x the distance from the sending end
L'	Inductance per unit length
m	Degree of shunt compensation
n	Turns ratio of an ideal transformer; number of sections in a cascade
P	Active power; P_{nat} natural power, P_{max} maximum transfer
Q	Reactive power
r_b	Equivalent radius of the bundle for the capacitance
R'	Resistance per unit length
s	Bundle spacing
S_{sc}	Short-circuit level at a terminal
U	Line-to-line voltage; $\underline{U}_S$, $\underline{U}_R$ at the terminals, U_n nominal
X	Series reactance of the path; X' per unit length, X_C of the series bank, X_{tot} of the whole path between the internal voltages
$\underline{Y}'$	Shunt admittance per unit length, $G' + j \cdot \omega \cdot C'$
$\underline{Z}'$	Series impedance per unit length, $R' + j \cdot \omega \cdot L'$
$\underline{T}$	Chain (ABCD) matrix of a two-port
$\underline{Z}_{eq}$	Series branch of the equivalent π; $\underline{Y}_{eq}$ its shunt branch
$\underline{Z}_c$	Image (characteristic) impedance of a two-port, $\sqrt{\underline{B}/\underline{C}}$
$\underline{Z}_{in}$	Impedance seen at port 1 of a terminated two-port
$\underline{Z}_L$	Impedance terminating a two-port
$\underline{Z}_w$	Wave (surge) impedance, $\sqrt{\underline{Z}'/\underline{Y}'}$
α	Attenuation constant
β	Phase constant; $\beta \cdot l$ the electrical length
$\underline{\gamma}$	Propagation constant, $\alpha + j \cdot \beta$
δ	Transmission angle between the terminal voltages
ε_0	Permittivity of free space
μ_0	Permeability of free space
$\underline{\theta}$	Image transfer constant of a two-port; $\underline{\gamma} \cdot l$ for a line
φ	Power-factor angle
ω	Angular frequency, $2 \cdot \pi \cdot f$

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3 Introduction

This document collects design notes for VSC HVDC transmission. Each topic is worked out here first, with its derivations, assumptions and sources; the corresponding slides of the HVDC design presentation are then derived from it.

Section 4 covers the creepage distance of insulators for DC overhead lines: why pollution, not overvoltage, sets it, how the site pollution is measured, and how the measured pollution, the insulator type and the altitude lead to the creepage of a 400 kV DC line.

Section 6 describes the DC fault ride-through of a half-bridge MMC symmetrical monopole on the record of the Pugalur–Thrissur link: the specified recovery times, the switchgear and the recovery sequence, the commissioning and field records, fault zones and selectivity, the changeover to STATCOM operation, the ratings the recovery affects and the design alternatives.

The numerical values given are indicative ranges from the standards and published guidance. They are not agreed AmWatt design constants; a project value follows from the measured site pollution and the supplier's test data.

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4 Creepage Distance

4.1 Definition and reference voltage

The creepage distance is the shortest distance along the surface of an insulator between the two conductive parts that normally carry the operating voltage. On a DC overhead line it is set by pollution, not by overvoltage: a wetted pollution layer conducts, and an insulator with too little creepage can flash over at normal operating voltage.

The creepage is specified as a unified specific creepage distance (USCD) in mm/kV, referred to the pole-to-earth DC operating voltage [1]. The total creepage of an insulator string follows as

$$L = \mathrm{USCD} \cdot U_{dpe} \cdot C_a \tag{1}$$

with L the total creepage in mm, U_{dpe} the pole-to-earth DC operating voltage in kV and C_a the altitude correction of Section 4.6. There is no single recommended USCD: it follows from the site pollution severity and the insulator type [1], [2].

4.2 Pollution flashover and the dry-band arc

A pollution flashover is caused by wet pollution. CIGRE Working Group B4.45 describes the process in detail [3]. Two forms of pollution are distinguished [4]: inland, the pollutants settle as dry dust and are wetted later by rain, fog or condensation (Type A); near the coast or at sea they arrive already wet, for example as salt fog (Type B). On an insulator with a hydrophilic surface the flashover then develops in six stages [3]:

1. **Pollution accumulation.** Pollutants from the air settle on the surface. How much accumulates depends on the amount in the air, on the electric field through electrostatic forces, and on the shed profile. The deposit may contain several kinds of salt.
2. **Wetting and dissolution.** Rain, fog, condensation or adsorption from humid air brings water to the deposit and dissolves at least part of it, so the layer becomes conductive. Different pollutants dissolve, and absorb water, at different rates.
3. **Leakage current.** The wet layer conducts, and a leakage current of a few mA to a few hundred mA flows along the surface.
4. **Dry-band formation.** The current heats the layer. Neither the pollution nor the water is evenly distributed, and the current density is highest near the pin and cap of a disc or on the narrow core of a long-rod or composite insulator, so some paths dry faster than others and non-conductive dry bands form.
5. **Partial arcs.** Nearly the full voltage of the section now stands across dry bands a few mm to a few cm wide. The field exceeds the strength of air, and partial arcs bridge the bands, burning in series with the remaining wet layer.
6. **Total flashover.** In the dynamic interplay of the dry-band arcs, one arc may extend across the whole insulator.

Two points follow [3]. A flashover needs both pollution and wetting: without pollution, or without the pollution becoming wet, none occurs. And both the accumulation and the flashover itself follow different patterns under AC and DC, Section 4.3.

The arc and the wet layer share the applied voltage. For an arc of length x on a creepage path of length L, the model of Obenaus [5] gives

$$U = x \cdot A \cdot I^{-n} + R' \cdot (L - x) \cdot I \tag{2}$$

where the first term is the arc voltage, whose gradient falls with rising current I (A and n are arc constants), and the second term is the voltage across the remaining wet layer of resistance R' per unit length. Figure 1(a) shows the arrangement.

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If the arc needs a lower voltage gradient than the wet layer it replaces, it elongates along the surface [6]. The current then rises, the arc gradient falls further, and the arc runs away until it bridges the whole insulator — the flashover. Otherwise it stalls, wanders and extinguishes, and appears only as surface sparking and leakage-current pulses. The creepage distance has to be long enough that the runaway condition is never reached at operating voltage.

4.3 Why DC needs more creepage than AC

The same site needs a markedly higher USCD on DC than on AC, for three reasons:

- **More pollution.** The steady DC field attracts charged particles, so a DC insulator collects roughly 1.5–2 times the pollution of an AC insulator at the same site [2].
- **No current zero.** An AC dry-band arc extinguishes at every current zero, 100 or 120 times per second, and has to re-ignite across a band that cools in the meantime. A DC arc burns continuously and can keep elongating without interruption, Figure 1(b).
- **Ionic migration.** Under DC, ions migrate through glass and porcelain and corrode the pin; DC insulators need DC-type glass and a zinc sleeve on the pin.

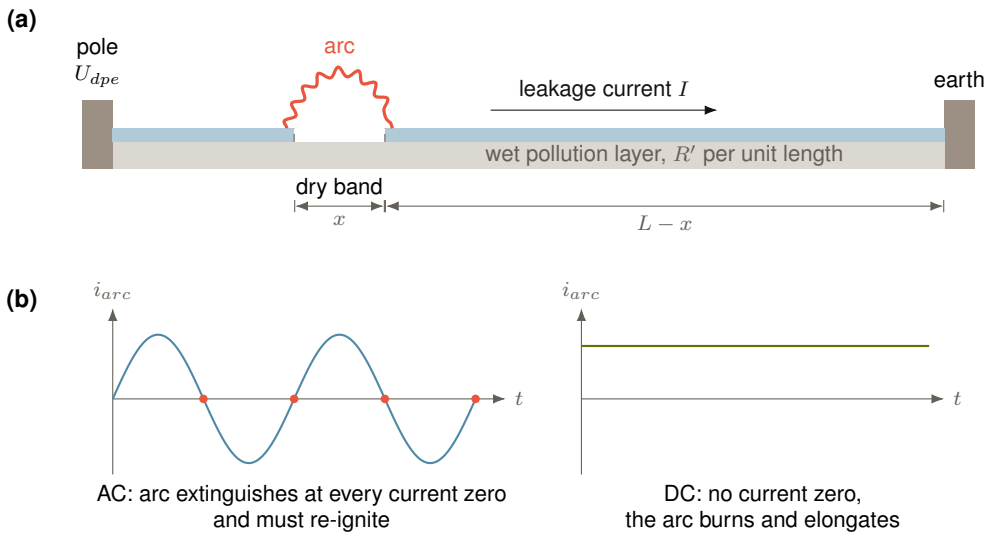


Figure 1: (a) Dry-band arc on the unrolled creepage path: a partial arc of length x bridges the dry band and burns in series with the wet layer of length $L - x$, Equation (2). (b) Arc current on AC and DC.

Table 1 summarises the differences. The composite (silicone) insulator counters the mechanism at its root: its water-repellent surface breaks the wet film into separate droplets, keeps the leakage current low and prevents dry bands from forming. Alternating large and small sheds with sufficient spacing also stop arcs from bridging between sheds and shortening the creepage path.

Aspect	AC	DC
Pollution build-up	wind and gravity	wind and gravity plus electrostatic attraction
Dry-band arc	extinguishes at every current zero	no current zero, continuous
Ionic migration in glass and porcelain	negligible	DC-type glass, zinc sleeve on the pin
Relative USCD	1	about 1.5–2

Table 1: Pollution performance of AC and DC insulators.

4.4 Site pollution severity: ESDD and NSDD

The pollution on an insulator has two parts. The *soluble deposit* is the salt — sea salt, industrial sulfates and nitrates, fertiliser, road de-icing salt, or the sodium, calcium and magnesium salts in desert and dry-lake dust — that dissolves into ions when wetted and makes the layer conduct. The *non-soluble deposit* — clay, sand, soot — does not conduct but holds water and keeps the soluble film wet for longer.

ESDD Equivalent salt deposit density: the mass of NaCl per surface area that, dissolved in water, gives the same conductivity as the soluble deposit washed off the insulator, in mg/cm^2 [4]. Converting to NaCl puts every salt on one scale, since different salts conduct differently per gram.

NSDD Non-soluble deposit density: the mass of the non-soluble residue per surface area, in mg/cm^2 [4].

Both are measured on reference insulators exposed on site for a year or more. A defined area A in cm^2 is washed with a volume V in cm^3 of demineralised water. The conductivity σ_θ of the wash water at temperature θ is corrected to 20 °C with the temperature factor b of [7], and converted into the salinity S_a in kg/m^3 (equal to mg/cm^3):

$$\sigma_{20} = \sigma_\theta \cdot [1 - b \cdot (\theta - 20\,^\circ\mathrm{C})] \tag{3}$$

$$S_a = (5.7 \cdot \sigma_{20})^{1.03} \tag{4}$$

$$\mathrm{ESDD} = \frac{S_a \cdot V}{A} \tag{5}$$

with σ in S/m. The wash water is then filtered; with W_i and W_f the dry mass of the filter in g before and after,

$$\mathrm{NSDD} = \frac{1000 \cdot (W_f - W_i)}{A} \tag{6}$$

Slowly soluble salts such as gypsum ($CaSO_4$), common in cement and some desert dusts, dissolve only partly; the measured ESDD then depends on the water volume and stirring time, so the wash procedure has to be fixed and stated with the results.

Together, ESDD and NSDD give the SPS class of the site, from very light to very heavy [4]. For DC, the SPS must be measured on DC-energised insulators, or estimated from AC or non-energised measurements with a DC/AC accumulation ratio of about 1.5–2 [1], [2].

The flashover strength per unit creepage falls with both quantities as a power law,

$$E_f = \frac{U_f}{L} = A_f \cdot \mathrm{ESDD}^{-a} \cdot \mathrm{NSDD}^{-b_n} \tag{7}$$

with A_f a constant of the insulator type. The required USCD therefore scales as

$$\mathrm{USCD}_2 = \mathrm{USCD}_1 \cdot \left(\frac{\mathrm{ESDD}_2}{\mathrm{ESDD}_1}\right)^{a} \cdot \left(\frac{\mathrm{NSDD}_2}{\mathrm{NSDD}_1}\right)^{b_n} \tag{8}$$

Typical exponents are $a \approx 0.2$–0.4 and $b_n \approx 0.1$–0.2, so the salt dominates: doubling the ESDD with $a = 0.3$ needs $2^{0.3} \approx 1.23$, i.e. 23 % more creepage. The exponents depend on insulator type and profile and are taken from the supplier's DC pollution tests or published data [2].

4.5 Indicative USCD values

Table 2 gives indicative USCD ranges for DC OHL insulators. Composite insulators need roughly 20–30 % less creepage than ceramic or glass, which is why most recent HVDC lines use composite long-rod insulators. The ranges are a starting point; the design value comes from the measured SPS.

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Site pollution	Ceramic / glass	Composite
Light	30–40 mm/kV	25–30 mm/kV
Medium	40–50 mm/kV	30–38 mm/kV
Heavy	50–60 mm/kV	38–45 mm/kV
Very heavy	mostly too long	45–55 mm/kV

Table 2: Indicative USCD for DC OHL insulators [1], [2].

4.6 Altitude correction

At lower air pressure the dry-band arc needs a lower gradient, so the pollution flashover voltage falls and more creepage is needed. Pollution tests show a power law in the pressure ratio, and the pressure falls exponentially with the altitude H [8], [9]:

$$U_f = U_{f0} \cdot \left(\frac{p}{p_0}\right)^m \qquad \frac{p}{p_0} = e^{-H/8150\,\mathrm{m}} \tag{9}$$

with U_f and U_{f0} the pollution flashover voltage at pressure p and at sea level p_0. Since the required creepage is inversely proportional to the flashover voltage, the creepage has to be multiplied by

$$C_a = e^{\,m \cdot \frac{H}{8150\,\mathrm{m}}} \tag{10}$$

This is the correction factor of IEC TS 60815-4 [1] and TB 518 [2], recommended for DC polluted insulators of both ceramic and polymeric material. CIGRE Working Group D1.44 reviews it in TB 705 [8], where the exponent is called n; it is written m here to keep it apart from the arc constant n of Equation (2).

4.6.1 Exponent m.

m is empirical: the slope of $\ln U_f$ over $\ln(p/p_0)$ from pollution tests at several pressures in a climate chamber or at high-altitude laboratories. It depends on the insulator type and material, the shed profile, the pollution level and the voltage type and polarity, and lies between 0 and 1 because only the arc depends on pressure, not the resistance of the wet layer. For DC, IEC TS 60815-4 and TB 518 suggest $m = 0.35$ until further information is available, a value still under consideration by CIGRE and IEC [8]. Since the correction may depend on the insulator characteristics and the pollution severity, it is applied per candidate insulator [1]. For comparison, IEC 60071-2 gives $m = 0.5$ for standard and up to 0.8 for anti-fog cap-and-pin insulators under AC [9].

Reference height. Two ways of counting the height are in use [8]:

(a) **Full height above sea level**, as in IEC 60071-2 and Equation (10). The USCD values are taken as sea-level values, and every site above sea level is corrected.

(b) **Height above 1000 m**, as in some industry practice. The USCD values are taken as valid up to 1000 m, and H is replaced by the height above 1000 m:

$$C_a = \begin{cases} 1 & H \le 1000\,\mathrm{m} \\ e^{\,m \cdot \frac{H - 1000\,\mathrm{m}}{8150\,\mathrm{m}}} & H > 1000\,\mathrm{m} \end{cases} \tag{11}$$

Variant (a) is the more conservative: it already adds 4.4 % at 1000 m, and above 1000 m it stays higher than variant (b) by the constant factor $e^{0.35 \cdot 1000/8150} = 1.044$. The specification has to state which variant applies, together with m. Table 3 compares both for $m = 0.35$, with the resulting creepage of a 400 kV ceramic string at medium pollution ($\mathrm{USCD} = 45\,\mathrm{mm/kV}$, so $L = 18.0\,\mathrm{m}$ before correction).

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	(a) Full height		(b) Above 1000 m	
Altitude H	C_a	L_{cer}	C_a	L_{cer}
1000 m	1.044	18.8 m	1.000	18.0 m
1500 m	1.067	19.2 m	1.022	18.4 m
2000 m	1.090	19.6 m	1.044	18.8 m
2500 m	1.113	20.0 m	1.067	19.2 m
3000 m	1.138	20.5 m	1.090	19.6 m

Table 3: Altitude correction C_a for $m = 0.35$ and the resulting creepage of the 400 kV ceramic string ($L = 18.0\,\mathrm{m}$ uncorrected).

TB 417 treats the altitude correction of the external insulation of outdoor DC yard equipment in the converter station [3, Section 7.1.8.3]:

- The air clearances of the DC switchyard are corrected for altitudes above 1000 m. The air clearances of the DC yard apparatus and the applicable test voltages are corrected above 1000 m according to the related equipment standards.
- For the creepage distance there is no generally accepted rule on altitude correction in the current international standards.
- Although the lower air density at high altitude may affect the insulation capability along the equipment surface, it is usual industrial practice not to correct creepage distances for altitude, in AC and DC substations alike.
- TB 417 considers it acceptable to apply the same design method to 800 kV DC equipment, i.e. no altitude correction of the creepage distance, as long as the specific creepage distance is selected conservatively and composite material is used for the equipment housing.

These statements concern station equipment. For the insulators of the DC overhead line, IEC TS 60815-4 and TB 518 recommend the correction C_a [1], [2], and this report applies it with Equation (10) or (11).

4.7 Design procedure

1. Determine the SPS along the route: ESDD and NSDD from test insulators exposed under DC voltage, or from service experience [1], [2].
2. Convert the SPS into a USCD with the IEC TS 60815-4 and CIGRE curves, or with the supplier's DC pollution test data for the chosen insulator profile.
3. Apply the altitude correction C_a, Equation (10) or (11), with m and the height convention stated in the specification, and the corrections for insulator diameter and profile.
4. Check the resulting string against the switching and lightning impulse requirements; the longer of the two governs.

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5 AC Power Transmission

The familiar statement that active power flows on the angle between two voltages, $P = U_S \cdot U_R / X \cdot \sin\delta$, is a special case. It holds for a short, lossless line, and its error grows with length.

This chapter builds the representation from the line parameters upwards and then applies it. Section 5.1 gives the general form: the four per-unit-length quantities, the two-port that follows from them, the exact power transfer relation, the conditions under which it collapses to the simple form, and the distributed solution that replaces the lumped one on long lines. Sections 5.2 and 5.3 give the parameters of an overhead line and of a cable, and Section 5.4 carries the chain through on one worked example. Sections 5.5 to 5.8 apply the representation: the natural power, the three transmission limits, the angular stability margin and the effect of series compensation.

5.1 Fundamentals

Nothing in this section distinguishes an overhead line from a cable. The two differ only in the values of the four parameters and in what those values make critical; Section 5.2 covers the overhead line and Section 5.3 the cable.

5.1.1 Line parameters

A transmission line is described by four parameters per unit length, two in series with the conductor and two in shunt to the return path. Each corresponds to one physical effect, Table 4.

	Parameter	Physical effect
R'	Resistance	Ohmic loss in the conductors
L'	Inductance	Magnetic field around the conductors
G'	Conductance	Leakage through the insulation and corona
C'	Capacitance	Electric field to the other conductors and to earth

Table 4: Line parameters per unit length

They enter as a series impedance and a shunt admittance per unit length,

$$\underline{Z}' = R' + j \cdot \omega \cdot L' \qquad \underline{Y}' = G' + j \cdot \omega \cdot C' \tag{12}$$

R' and L' follow from the conductor itself and from the arrangement of the conductors; G' and C' from the insulation between them. The expressions differ between an overhead line and a cable and are given in Sections 5.2 and 5.3.

Taken over a finite length the four parameters arrange themselves as the nominal π-section of Figure 2: the series branch carries R' and L', and the shunt branch G' and C' is split equally between the two ends.

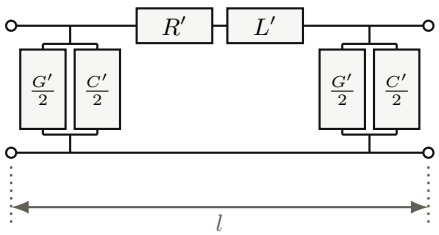


Figure 2: Nominal π-section of a line of length l, written in the per-unit-length parameters

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5.1.2 The line as a two-port

Over a length l the parameters give a series impedance and a shunt admittance

$$\underline{Z} = \underline{Z}' \cdot l \qquad \underline{Y} = \underline{Y}' \cdot l \tag{13}$$

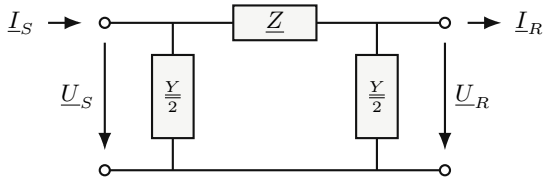


Figure 3: The same π-section as a two-port: the whole series impedance $\underline{Z}$ between the rails, half the shunt admittance $\underline{Y}$ at each end

which the nominal π arranges as in Figure 3: the whole series impedance between the rails and half the shunt admittance at each end. This is the standard two-port description of a linear network, of which the line is one case; Appendix A gives the general form for the reader who wants it, and is not needed here. The ports are named by function throughout this section, the sending end and the receiving end. The line is then a two-port,

$$\begin{bmatrix} \underline{U}_S \\ \underline{I}_S \end{bmatrix} = \begin{bmatrix} \underline{A} & \underline{B} \\ \underline{C} & \underline{D} \end{bmatrix} \cdot \begin{bmatrix} \underline{U}_R \\ \underline{I}_R \end{bmatrix} \tag{14}$$

with, for the nominal π,

$$\underline{A} = \underline{D} = 1 + \frac{\underline{Z} \cdot \underline{Y}}{2}, \qquad \underline{B} = \underline{Z}, \qquad \underline{C} = \underline{Y} \cdot \left(1 + \frac{\underline{Z} \cdot \underline{Y}}{4}\right) \tag{15}$$

The shunt admittance appears in $\underline{A}$. A line with no shunt capacitance has $\underline{A} = 1$ and behaves as a single series impedance; the departure of $|\underline{A}|$ from unity measures the departure from that case.

5.1.3 Power transfer

Solving (14) for the receiving-end current gives

$$\underline{I}_R = \frac{\underline{U}_S - \underline{A} \cdot \underline{U}_R}{\underline{B}} \tag{16}$$

Taking the receiving-end voltage as the phase reference, so that δ is the angle by which the sending-end voltage leads it,

$$\underline{U}_S = U_S \cdot e^{j \cdot \delta}, \quad \underline{U}_R = U_R, \quad \underline{A} = |A| \cdot e^{j \cdot \vartheta_A}, \quad \underline{B} = |B| \cdot e^{j \cdot \vartheta_B} \tag{17}$$

the receiving-end complex power $\underline{S}_R = \underline{U}_R \cdot \underline{I}_R^*$ separates into

$$P_R = \frac{U_S \cdot U_R}{|B|} \cdot \cos\left(\vartheta_B - \delta\right) - \frac{|A| \cdot U_R^2}{|B|} \cdot \cos\left(\vartheta_B - \vartheta_A\right) \tag{18}$$

$$Q_R = \frac{U_S \cdot U_R}{|B|} \cdot \sin\left(\vartheta_B - \delta\right) - \frac{|A| \cdot U_R^2}{|B|} \cdot \sin\left(\vartheta_B - \vartheta_A\right) \tag{19}$$

with U_S and U_R line-to-line magnitudes, so that P_R and Q_R are three-phase quantities.

Two numbers carry the whole behaviour of the line. The angle ϑ_B sets where the maximum of P_R falls: differentiating (18) puts it at $\delta = \vartheta_B$, which is 90° only for a lossless line. The magnitude $|A|$ scales the second term, a constant offset independent of δ that exists only because the line charges itself.

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5.1.4 The short, lossless line

For a short line the shunt admittance vanishes and $\underline{A} = \underline{D} = 1$, so $|A| = 1$ and $\vartheta_A = 0$. For a lossless line $\underline{B} = \underline{Z} = j \cdot X$, so $|B| = X$ and $\vartheta_B = 90°$. With $\cos(90° - \delta) = \sin\delta$ and $\cos 90° = 0$, equations (18) and (19) collapse to

$$P = \frac{U_S \cdot U_R}{X} \cdot \sin\delta \qquad Q_R = \frac{U_S \cdot U_R \cdot \cos\delta - U_R^2}{X} \tag{20}$$

with the series reactance

$$X = \omega \cdot L' \cdot l \tag{21}$$

For small angles this separates further into $P \approx U_S \cdot U_R \cdot \delta / X$ and $Q_R \approx U_R \cdot (U_S - U_R) / X$: active power is carried by the angle between the terminal voltages, reactive power by the difference in their magnitude. The decoupling holds for small angles and is the basis for treating the two separately in estimation, although (20) is not accurate enough to compute with on a long line.

The X of (20) is not the line alone but the whole path between the two internal voltages — the source impedances and the transformers at both ends are in series with it — so the short-circuit level at the terminals belongs in any statement of transfer capability.

5.1.5 Distributed representation

The π-section lumps parameters that are distributed along the line. The distributed description uses two quantities,

$$\underline{\gamma} = \alpha + j \cdot \beta = \sqrt{\underline{Z}' \cdot \underline{Y}'} \qquad \underline{Z}_w = \sqrt{\frac{\underline{Z}'}{\underline{Y}'}} \tag{22}$$

the propagation constant and the wave impedance. Voltage and current are then defined at every cross-section of the line rather than only at its terminals, Figure 4. At a distance x from the sending end of a line of length l they are

$$\underline{U} = \underline{U}_R \cdot \cosh\left(\underline{\gamma} \cdot (l - x)\right) + \underline{Z}_w \cdot \underline{I}_R \cdot \sinh\left(\underline{\gamma} \cdot (l - x)\right) \tag{23}$$

$$\underline{I} = \frac{\underline{U}_R}{\underline{Z}_w} \cdot \sinh\left(\underline{\gamma} \cdot (l - x)\right) + \underline{I}_R \cdot \cosh\left(\underline{\gamma} \cdot (l - x)\right) \tag{24}$$

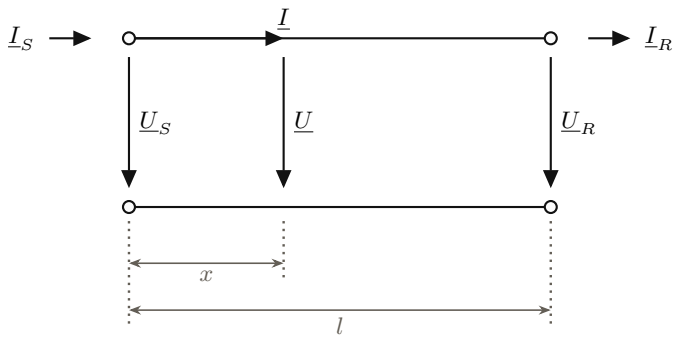


Figure 4: Terminal quantities of the two-port and the local voltage and current at a distance x along the line

Evaluating (23) and (24) at $x = 0$ and comparing with (14) identifies the exact two-port parameters,

$$\underline{A} = \underline{D} = \cosh\left(\underline{\gamma} \cdot l\right), \qquad \underline{B} = \underline{Z}_w \cdot \sinh\left(\underline{\gamma} \cdot l\right), \qquad \underline{C} = \frac{\sinh\left(\underline{\gamma} \cdot l\right)}{\underline{Z}_w} \tag{25}$$

which replace the nominal-π entries of (15). The relations (18) and (19) are unchanged; only the way $\underline{A}$ and $\underline{B}$ are computed differs.

The exact result can itself be written as a π. Matching the branches of a π-section to (25) entry by entry gives

$$\underline{Z}_{eq} = \underline{Z}_w \cdot \sinh\left(\underline{\gamma} \cdot l\right), \qquad \frac{\underline{Y}_{eq}}{2} = \frac{1}{\underline{Z}_w} \cdot \tanh\left(\frac{\underline{\gamma} \cdot l}{2}\right) \tag{26}$$

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the *equivalent* π, often called the long-line correction. It is not an approximation: one π with these two branches reproduces the distributed line exactly at its terminals at that frequency, because $1 + \sinh\left(\underline{\gamma} \cdot l\right) \cdot \tanh\left(\underline{\gamma} \cdot l/2\right) = \cosh\left(\underline{\gamma} \cdot l\right)$. The nominal π of (15) is its small-argument limit, with $\sinh u \to u$ and $\tanh u \to u$. This is what a load-flow program stores for a long line, and it allows a π-shaped model at any length. Section 5.4.3 puts numbers on both.

What decides whether the lumped form will do is the electrical length $\beta \cdot l$, not the length in km. Since $\beta = \omega \cdot \sqrt{L' \cdot C'}$ for a low-loss line, a cable reaches a given electrical length in a fraction of the distance an overhead line needs, and the lumped approximation fails correspondingly earlier.

5.2 Overhead Lines

5.2.1 Geometry and the line parameters

On an overhead line the conductors are bare and the insulation is air, so the inductance and the capacitance follow from the geometry of the tower and of the conductor bundle alone:

$$L' = \frac{\mu_0}{2 \cdot \pi} \cdot \ln \frac{D_m}{r_b} \qquad C' = \frac{2 \cdot \pi \cdot \varepsilon_0}{\ln\left(D_m / r_{b_c}\right)} \tag{27}$$

with D_m the geometric mean distance between the phases and r_b, r_{b_c} the equivalent radii of the bundle for the inductance and the capacitance respectively. The two differ because the inductance is referred to the geometric mean radius of the sub-conductors and the capacitance to their physical radius.

Both parameters depend on the geometry only through a logarithm, and on the same ratio D_m / r_b. That makes the geometry a weak lever: the phase spacing and the bundle radius have to be changed by a factor to move L' or C' by a fraction. Length, by contrast, enters linearly, so on an overhead line the series reactance of (21) is set by the route rather than by the tower.

The flat (horizontal) configuration used throughout this report is drawn to scale in Figure 5: three phases on one crossarm at a spacing D, so that the outer phases see $2 \cdot D$, each phase a bundle of three sub-conductors. Table 5 gives its dimensions. It gives the lowest towers and the widest right-of-way of the common arrangements.

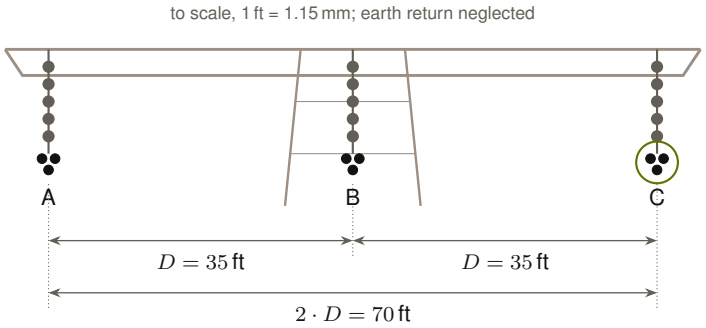


Figure 5: Tower head of the reference line, flat configuration; the marked phase is drawn in detail in Figure 6. Tower body and insulator lengths are schematic

Bundling is the geometric change with the largest electrical effect. Dividing the phase current between sub-conductors raises the equivalent radius r_b well above that of a single conductor, and with it reduces L'; it also reduces the surface gradient and so the corona loss. Table 6 shows the effect for one configuration: going from one sub-conductor to three removes a third of the reactance, while doubling r_b again — or halving D_m — removes only a further 15 %.

The shunt branch of an overhead line is effectively a pure capacitance. The conductance G' comprises the leakage across the insulator strings and the corona loss. In normal weather it is about a hundred times smaller than the susceptance $\omega \cdot C'$ and is set to zero in load-flow work. It is not negligible for loss studies in foul weather, where corona dominates it.

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Parameter		Value
D phase spacing	ft	35
$2 \cdot D$ outer phases	ft	70
Sub-conductors per phase		3
s bundle spacing	in	18
d conductor diameter	in	1.762
GMR of a sub-conductor	ft	0.0588

Table 5: Tower-head geometry of the reference line, Bluebird ACSR 2156 kcmil 84/19, single circuit [10]

Sub-conductors per phase	r_b	X'	relative
	m	Ω/km	%
1	0.018	0.499	148
2	0.091	0.377	112
3	0.155	0.336	100
4	0.222	0.310	92

Table 6: Effect of bundling on the series reactance, 35 ft flat configuration with 18 in bundle spacing at 60 Hz

Because the conductor spacing is large and the insulation is air, C' stays small. The wave impedance $\underline{Z}_w = \sqrt{\underline{Z}'/\underline{Y}'}$ is correspondingly high — a few hundred ohms — and the charging current per unit length is low enough that an overhead line can run for hundreds of kilometres before it has to be compensated.

5.2.2 Bundle and phase equivalents

The two radii of (27) and the mean phase distance follow from the tower head and the bundle. For a three-conductor bundle of spacing s made of sub-conductors of diameter d and geometric mean radius GMR, Figure 6, and a flat configuration of phase spacing D in which the outer phases see $2 \cdot D$,

$$GMR_b = \sqrt[3]{GMR \cdot s^2}, \qquad r_b = \sqrt[3]{\frac{d}{2} \cdot s^2}, \qquad GMD = \sqrt[3]{D \cdot D \cdot 2 \cdot D} = D \cdot \sqrt[3]{2} \tag{28}$$

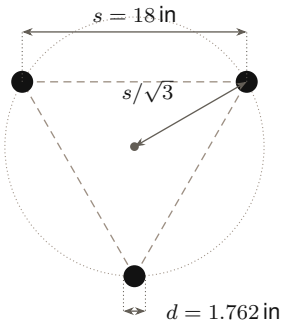


Figure 6: Three-conductor bundle of the reference line, apex down, with the circle of radius $s/\sqrt{3}$ on which the sub-conductors sit

with GMR_b the radius entering L' and r_b the one entering C'. For a two-conductor bundle the same construction gives $GMR_b = \sqrt{GMR \cdot s}$ and $r_b = \sqrt{d/2 \cdot s}$. The GMD above assumes a transposed line.

Table 7 evaluates them for the reference tower head of Section 5.4 — 35 ft flat, 18 in bundle spacing, Bluebird ACSR 2156 kcmil with $d = 1.762$ in and $GMR = 0.0588$ ft — with three and with two sub-conductors per phase.

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Quantity		$3\times$	$2\times$
GMR_b	in	6.11	3.56
r_b	in	6.58	3.98
GMD	ft	44.1	44.1
L'	mH/km	0.892	1.000
C'	nF/km	12.68	11.38
$Z_w = \sqrt{L'/C'}$	Ω	265	296

Table 7: Bundle equivalents and the resulting parameters, three against two sub-conductors per phase

The third sub-conductor lowers Z_w by about 10 % and so raises the natural power of Section 5.5 by about 11 %. A flat configuration gives the lowest towers and the widest right-of-way; the outer phases see twice the phase spacing, which is what the cube root in (28) averages.

5.2.3 Conductor codewords

A conductor size does not define a conductor. ACSR of 795 kcmil exists in four standard strandings — Cuckoo 24/7, Drake 26/7, Mallard 30/19 and Condor 54/7 — each with a different steel content and therefore different strength, weight and sag. The codeword fixes area, stranding and core in one word.

The names began as Alcoa product nomenclature in the 1920s and 1930s and were later adopted into the ASTM standards [13]. The family identifies the construction: ACSR and ACSS take bird names (Drake, Falcon, Rail, Cardinal), AAC flowers (Rose, Iris, Tulip, Peony) [14] and AAAC US city names (Akron, Amherst, Anaheim) [15]. The name carries no size information — Bluebird is 2156 kcmil, Condor only 795.

Other systems use the same idea with different animals. BS 215 and Indian practice use mammals: Dog, Wolf, Lynx, Panther, Zebra, Moose [16]. A 400 kV Indian line specified as “twin Moose” is the equivalent of a North American “bundled Drake”. Metric designs in continental Europe usually drop codewords altogether and quote the aluminium and steel areas directly, for example 490/65 mm^2.

The practical consequence is that a conductor has to be quoted with its standard as well as its codeword, since the same name can appear in more than one national system.

5.3 Cables

A cable carries the same four parameters and the same representation, but the geometry inverts the shunt branch. Conductor and screen are separated by a few centimetres of solid dielectric rather than by metres of air, so the capacitance per unit length rises by more than an order of magnitude while the inductance falls, Table 8.

Parameter		OHL	Cable
R'	Ω/km	0.1–1	0.1–1
L'	mH/km	≈ 1	0.31–1.42
G'	µS/km	0.05	≈ 1
C'	nF/km	≈ 10	180–420

Table 8: Indicative per-unit-length parameters, overhead line against XLPE cable [17]

Three consequences follow from these values. Together they make a cable a different design problem from a shorter overhead line.

Low wave impedance. With C' up and L' down, $\underline{Z}_w$ falls from a few hundred ohms to under a hundred. For a 230 kV XLPE cable with $L' = 1.35$ mH/km and $C' = 190$ nF/km, $\underline{Z}_w = 84\,\Omega$ against $266\,\Omega$ for the

overhead line of Section 5.4. The natural power rises in proportion, but it is not a loading that is reached: the charging current binds first.

High charging power. The charging reactive power per unit length is $Q'_c = U^2 \cdot \omega \cdot C'$. For the same example cable $\omega \cdot C' = 71.6\,\mu$S/km, giving 3.79 MVAr/km and a charging current of 9.5 A/km at 230 kV. That current flows whether or not the cable is loaded, it is drawn from both ends, and it consumes conductor ampacity that is then unavailable for load current. Beyond a critical length the charging current occupies the whole ampacity. Shunt reactors at the ends — and, for long routes, at intermediate points — are therefore a condition of operation.

Short electrical length. The phase constant $\beta = \omega \cdot \sqrt{L' \cdot C'}$ rises with the capacitance, so a cable reaches a given $\beta \cdot l$ in a fraction of the route length an overhead line needs. The example cable has a wavelength of 1041 km against 4936 km for the reference overhead line, so one kilometre of this cable has the electrical length of 4.7 km of that line. The lumped π of Section 5.1.2 therefore loses accuracy at a few tens of kilometres rather than a few hundred, and the distributed form of (25) applies at correspondingly shorter routes.

Where intermediate compensation is impossible — a submarine crossing, or a long cable with no accessible landing point — these three together rule out AC. The comparison is then between an AC and a DC scheme rather than between a cable and an overhead line.

5.4 Worked Example: the 500 kV Reference Line

Most of the numbers in this report come from one line. It is a 500 kV, 60 Hz overhead line of 300 mi (483 km) in flat configuration, with three Bluebird ACSR 2156 kcmil sub-conductors per phase, 35 ft phase spacing and 18 in bundle spacing. The conductor data are the Southwire ACSR tables [10] and the resistance is the value at 75 °C. Table 9 gives the line and the parameters that follow from it by (27). Section 5.5 uses a second line — 400 kV, 50 Hz, 400 km — and says so where it does.

Quantity		Value
U_n	kV	500
f	Hz	60
l	mi (km)	300 (483)
Conductor		$3\times$ Bluebird ACSR
Geometry	ft / in	35 flat / 18 bundle
R' (75 °C)	mΩ/km	56.9
L'	mH/km	0.891
$X' = \omega \cdot L'$	Ω/km	0.336
G'	µS/km	0.05
C'	nF/km	12.7

Table 9: 500 kV reference line and its per-unit-length parameters

At 60 Hz the susceptance is $\omega \cdot C' = 4.79\,\mu$S/km, about a hundred times the conductance, which is the overhead-line case of Section 5.2: the shunt branch is treated as a pure capacitance throughout.

5.4.1 Line constants

Table 10 gives the lumped and distributed constants of the whole 300 mi line.

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Quantity		Value
$\underline{Z}$	Ω	$27.5 + j\,162.2$
$\underline{Y}$	mS	$j\,2.31$
Charging at 1.0 p.u.	MVAr	578
α	Np/km	1.07×10^{-4}
β	rad/km	1.27×10^{-3}
$\beta \cdot l$	°	35.2
Wavelength	km	4936
$\underline{Z}_w$	Ω	$266 - j\,22$
$P_{nat} = U_n^2/\mathrm{Re}\{\underline{Z}_w\}$	MW	939

Table 10: Computed quantities of the reference line at 300 mi

5.4.2 Deviation of the nominal π

Table 11 evaluates all four two-port parameters of this line at several lengths, once as the nominal π of (15) and once exactly from (25); $\underline{D} = \underline{A}$ in both. Table 12 gives the deviations, and Figure 7(a) the same comparison continuously against length [12].

The four parameters do not deviate equally.

- $|\underline{B}|$ deviates most, and high: the nominal π holds the whole series impedance in one branch, while on the line it is distributed among the shunt capacitance. P_{max} varies inversely with $|\underline{B}|$, so the nominal π understates the transfer capability.
- $|\underline{A}|$ and $|\underline{C}|$ deviate low. $|\underline{A}|$ stays within 1 % to 350 mi, at which length $|\underline{B}|$ deviates by 9.0 %; a model assessed on its voltage ratio alone therefore gives no indication of the error in its series parameter.
- P_{max} carries both deviations and lies between them: -7.2 % at 300 mi and -25.8 % at 600 mi.

The deviation of $|\underline{B}|$ passes 1 % at 120 mi, 6.5 % at 300 mi and 12.0 % at 400 mi.

l	$\beta \cdot l$	$\lvert\underline{A}\rvert$		$\lvert\underline{B}\rvert$		$\lvert\underline{C}\rvert$	
mi	°	π	exact	π, Ω	exact, Ω	π, mS	exact, mS
100	11.7	0.979	0.979	54.8	54.5	0.762	0.765
200	23.5	0.917	0.918	109.7	106.7	1.477	1.499
300	35.2	0.813	0.819	164.5	154.4	2.095	2.170
400	46.9	0.669	0.686	219.4	195.8	2.570	2.751
600	70.4	0.280	0.351	329.1	252.9	2.904	3.553

Table 11: Two-port parameters of the reference line against length, nominal π and exact; $\underline{D} = \underline{A}$ throughout [12]

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l	$\beta \cdot l$	$\lvert\underline{A}\rvert$	$\lvert\underline{B}\rvert$	$\lvert\underline{C}\rvert$	P_{max}
mi	°	%	%	%	%
100	11.7	-0.0	$+0.7$	-0.4	-0.8
200	23.5	-0.1	$+2.8$	-1.4	-3.3
300	35.2	-0.7	$+6.5$	-3.4	-7.2
400	46.9	-2.5	$+12.0$	-6.6	-12.4
600	70.4	-20.0	$+30.1$	-18.2	-25.8

Table 12: Deviation of the nominal π from the exact two-port, and the resulting error in the maximum transfer at 500 kV [12]

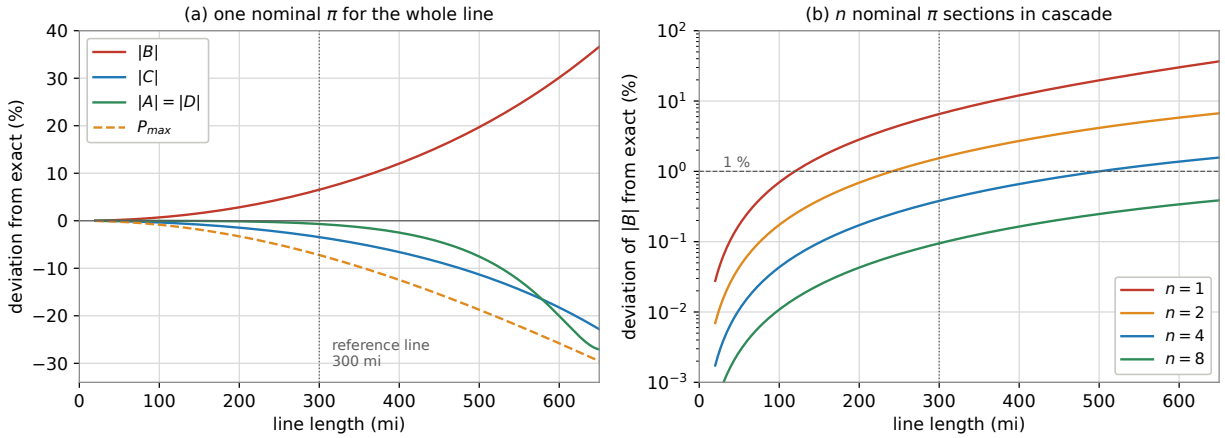


Figure 7: (a) Deviation of the nominal π against line length; (b) the same deviation of $\lvert\underline{B}\rvert$ when the line is built from n cascaded π-sections. Both stop short of the quarter wavelength, where $\lvert\underline{A}\rvert$ passes through zero

5.4.3 Cascading shorter sections

One lumped section cannot represent a distributed parameter. A second approach divides the line into n pieces of length l/n, models each as its own nominal π, and connects the two-ports in cascade. The chain matrices multiply, so the whole line is

$$\underline{T} = \left[\underline{T}_{\pi}\left(\frac{l}{n}\right)\right]^{n} \tag{29}$$

a single matrix power rather than n separate models. Each section covers a shorter length and therefore deviates less, and as $n \to \infty$ the product tends to the distributed solution of (25).

Table 13 and Figure 7(b) give the convergence. Once the sections are electrically short the deviation falls with $1/n^2$, so doubling n divides it by four. The deviation is set by the electrical length of a *section* and not by that of the line: on this conductor, sections of about 120 mi hold $\lvert\underline{B}\rvert$ within 1 % at any total length, the same length that Table 11 gives for an undivided line.

n	**section**	deviation of $\lvert\underline{B}\rvert$, %				
		100 mi	200 mi	300 mi	400 mi	600 mi
1	l	$+0.70$	$+2.83$	$+6.53$	$+12.03$	$+30.13$
2	$l/2$	$+0.17$	$+0.69$	$+1.54$	$+2.71$	$+5.81$
4	$l/4$	$+0.04$	$+0.17$	$+0.38$	$+0.66$	$+1.38$
8	$l/8$	$+0.01$	$+0.04$	$+0.09$	$+0.16$	$+0.34$
16	$l/16$	$+0.00$	$+0.01$	$+0.02$	$+0.04$	$+0.08$

Table 13: Nominal π-sections in cascade: deviation of $\lvert\underline{B}\rvert$ from the exact two-port against the number of sections [12]

The equivalent π of (26) reaches the same result with one section and no deviation at any length. On this

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line at 300 mi it moves the series branch from $27.5 + j\,162.2\,\Omega$ to $24.1 + j\,152.5\,\Omega$ and the shunt branch from 1.156 to 1.193 mS at each end; at 600 mi from $54.9 + j\,324.4\,\Omega$ to $30.4 + j\,251.0\,\Omega$ and from 2.312 to 2.650 mS. Sixteen cascaded nominal sections still deviate by 0.08 % on $|\underline{B}|$ and require sixteen matrix products.

Cascading applies where a single equivalent π does not.

- **Something is connected part-way along.** A midpoint series bank is a two-port cascaded between two half-lines, Section 5.8, and a tapped substation or a transposition point divides the line in the same way. The equivalent π describes a line between its own two ends and nothing else.
- **Quantities along the line are wanted.** The voltage profiles of Section 5.5.3 are the intermediate port quantities of a cascade; a single two-port gives only its terminals.
- **The line is not uniform.** Different conductor, spacing or bundle over part of the route gives each part its own matrix, and the product handles it with no extra theory. The equivalent π assumes one $\underline{\gamma}$ and one $\underline{Z}_w$ over the whole length.
- **A time-domain or frequency-sweep model is wanted.** The cascade is a lumped circuit a simulator can integrate directly, which is why electromagnetic-transient models still use chained sections rather than a closed-form $\cosh$.

Where the line is uniform and only its terminals matter, the equivalent π is the right model, and the cascade is a way of understanding where it comes from. Appendix A.4 sets the cascade beside the other ways two-ports combine, which are not interchangeable with it.

5.4.4 Comparison of the three representations

Table 14 evaluates the two-port parameters and the crest of the power-angle curve for the same line under the three representations of Section 5.1. The simplification of (20) moves the maximum from $81.0°$ to $90°$ and overstates it by 13 %; the nominal π keeps the angle but understates the maximum by 7 %, because its $|\underline{B}|$ is too large. At 300 mi $|\underline{A}| = 0.82$, 18 % below the unity that the single series impedance of (20) assumes. The numbers follow from the model of Section 5.1 implemented in [11].

Model	$\lvert A\rvert$	ϑ_B	$\lvert B\rvert$	P_{max}
		°	Ω	MW
Short, lossless	1.000	90.0	162.2	1541
Nominal π	0.813	80.4	164.5	1266
Distributed	0.819	81.0	154.4	1364

Table 14: Two-port parameters and maximum transfer at 300 mi, both terminals at 1.0 p.u.

5.4.5 The surrounding system

The reactance that appears in (20) is the whole path between the two internal voltages, not the line:

$$X_{tot} = X_{th_s} + X_{tr_s} + X' + X_{tr_r} + X_{th_r}, \qquad X_{th} = \frac{U_n^2}{S_{sc}}, \qquad \mathrm{SCR} = \frac{S_{sc}}{P_n} \tag{30}$$

with X_{th} the Thevenin reactance of the system behind a terminal, X_{tr} that of the transformer and S_{sc} the short-circuit level there. Taking the lossless $P_{max} = U_n^2/X_{tot}$ with equal systems at both ends and the transformers neglected, Table 15 shows how much of the capability belongs to the terminals rather than to the line.

A quarter of the capability of this line into 10 GVA systems is lost to the terminals rather than to the line itself, which is why a loadability figure quoted for a line in isolation overstates it. The same quantity, normalised as the short-circuit ratio, is what governs converter performance on a DC link.

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SCL, each end	X_{th}	X_{tot}	P_{max}
GVA	Ω	Ω	MW
5	50.0	253.4	986
10	25.0	203.4	1229
20	12.5	178.4	1401
stiff	0.0	153.4	1630

Table 15: Effect of the surrounding system on the maximum transfer of the 300 mi reference line, $X' = 153.4\,\Omega$, transformers neglected

5.4.6 Scope of P_{max}

P_{max} is not a transfer capability. It is the crest of the power-angle curve of an uncompensated line between two stiff systems, at an angle outside the operating range. The transfer the line can carry follows from the thermal rating of the conductors, the permissible voltage drop and the angular stability margin, evaluated on the contingency network rather than the intact one. Section 5.6 covers those limits; the representation of Section 5.1 is what they are evaluated on.

5.5 Natural Power

5.5.1 Definition

A line terminated in its own wave impedance carries its *natural power*: the reactive power generated by the shunt capacitance exactly cancels the reactive power consumed by the series inductance, the line exchanges no reactive power with either end, and its voltage magnitude stays constant along the route. North American practice calls the same quantity the surge-impedance loading, SIL.

On an overhead line $\omega \cdot L' \gg R'$ and $\omega \cdot C' \gg G'$, so the wave impedance of (22) is within about a percent of the real quantity $\sqrt{L'/C'}$, and it is that real value the natural power is quoted from:

$$Z_w \approx \sqrt{\frac{L'}{C'}} \qquad P_{nat} = \mathrm{SIL} = \frac{U^2}{Z_w} \tag{31}$$

Z_w is a property of the conductor geometry alone. It does not depend on the length of the line, and in the lossless form not on frequency either. P_{nat} is therefore a reference rather than a rating: below it the line is a net source of reactive power, above it a net sink.

The same result follows from the two-port equations. Closing the line with $\underline{Z}_w$ makes $\underline{I}_R = \underline{U}_R/\underline{Z}_w$, and (23) and (24) collapse to a single travelling wave with no reflection,

$$\underline{U}(x) = \underline{U}_R \cdot e^{\underline{\gamma} \cdot x}, \qquad \underline{I}(x) = \underline{I}_R \cdot e^{\underline{\gamma} \cdot x} \tag{32}$$

so that $\underline{U}/\underline{I} = \underline{Z}_w$ everywhere and the magnitude falls only by the attenuation $e^{-\alpha \cdot l}$, Figure 8. Only the complex $\underline{Z}_w$ matches perfectly; terminating in the real $\sqrt{L'/C'}$ leaves a small residual reflection.

5.5.2 The reactive balance

Read as a reactive power balance, a line of length l at line-to-line voltage U carrying phase current I generates and consumes

$$Q_C = \omega \cdot C' \cdot l \cdot U^2 \qquad Q_L = 3 \cdot \omega \cdot L' \cdot l \cdot I^2 \tag{33}$$

Natural loading is where the two cancel. Setting $Q_C = Q_L$ eliminates both l and ω and gives $I = U/\left(\sqrt{3} \cdot Z_w\right)$, hence $P_{nat} = \sqrt{3} \cdot U \cdot I = U^2/Z_w$ again. The cancellation is independent of length, which is why the natural power is a property of the cross-section and not of the route.

The worked line of this section is a 400 kV, 50 Hz line of 400 km with a two-conductor bundle, $R' = 25\,\mathrm{m\Omega/km}$, $X' = 0.30\,\Omega/\mathrm{km}$ and $C' = 12.5\,\mathrm{nF/km}$, giving $L' = 0.955\,\mathrm{mH/km}$, $Z_w = 276.4\,\Omega$, $|\underline{Z}_w| = 276.9\,\Omega$, $\beta \cdot l = 24.88°$ and $P_{nat} = 579\,\mathrm{MW}$. It is not the 500 kV line of Section 5.4.

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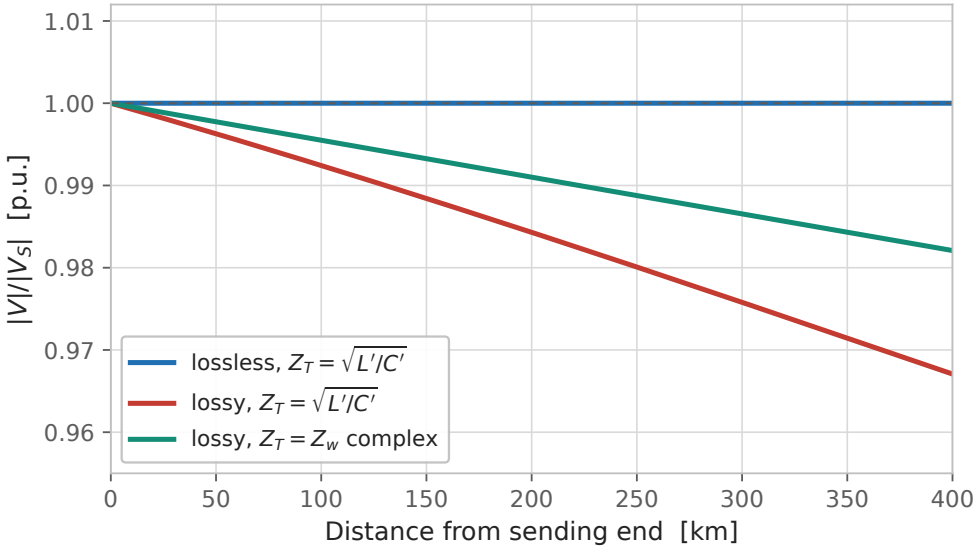


Figure 8: Voltage along a line closed in its wave impedance: the profile is flat apart from the attenuation, and the residual ripple of the real-valued termination

Losses move the balance. On the lossless line the zero crossing of the net reactive flow reproduces U^2/Z_w to six decimals; on the real line it falls to 566 MW, 2.2 % below. The sending-end current carries the load plus the losses, so $3 \cdot I^2 \cdot X' \cdot l$ is larger and the balance is reached at a lower transfer. Figure 9 plots the net reactive flow of both against the transfer.

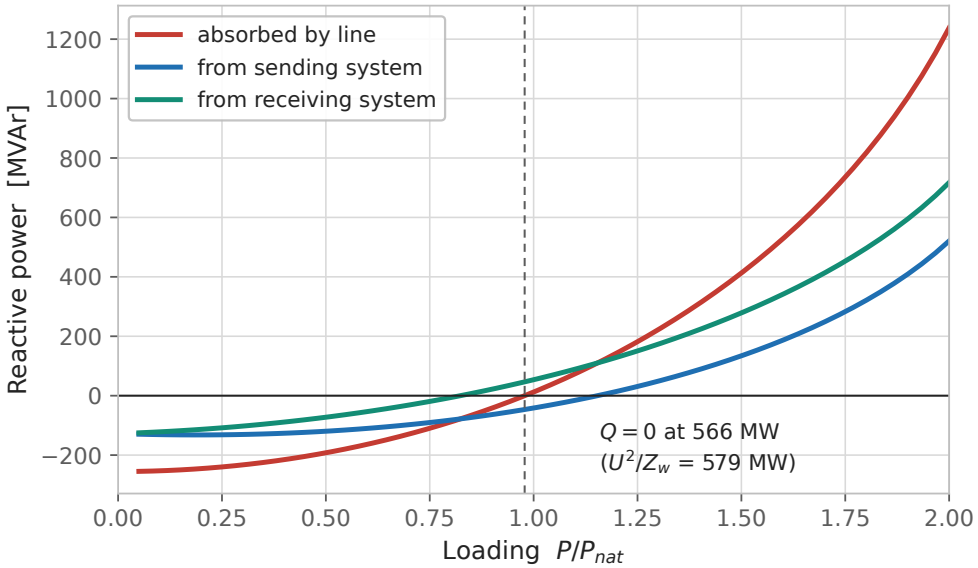


Figure 9: Net reactive power absorbed by the line against transfer, $Q_{line} = \Im\{\underline{S}_S\} - \Im\{\underline{S}_R\}$, lossless and real line

5.5.3 Voltage profile

With both terminals held at 1.0 p.u. and the transmission angle solved for each loading, the profile along the line shows the balance directly, Figure 10. Below P_{nat} the shunt capacitance dominates and the middle of the line rises, to 1.024 p.u. at no load. At P_{nat} the profile is flat to four decimals. Above it the series reactance dominates and the middle sags, to 0.873 p.u. at $2 \cdot P_{nat}$, by which point the angle across the line has reached $63°$.

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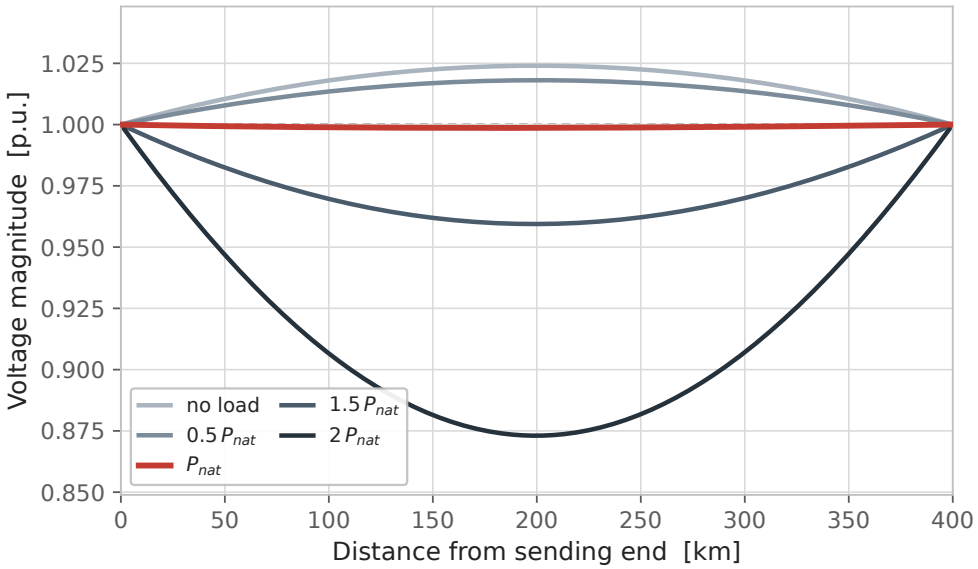


Figure 10: Voltage profile along the 400 km line at several loadings, both terminals at 1.0 p.u.

5.5.4 Typical values

Table 16 gives the natural power of common line types.

Line	U	C'	Z_w	P_{nat}	S_{th}/P_{nat}
	kV	nF/km	Ω	MW	
230 kV, $1\times$ Drake	230	9.0	376.1	141	2.8
345 kV, $2\times$ bundle	345	11.8	292.3	407	2.6
400 kV, $2\times$ bundle	400	12.5	276.4	579	2.4
500 kV, $3\times$ bundle	500	13.5	256.6	974	2.3
765 kV, $4\times$ bundle	765	13.5	250.8	2334	2.0

Table 16: Natural power of typical line types

Three things follow. Z_w spans only 251 to 376 Ω across the whole range while P_{nat} spans 141 to 2334 MW, because P_{nat} is a U^2 quantity: voltage, not geometry, sets the transfer. Bundling and tighter phase spacing are the only levers on Z_w, and they work on both terms, lowering L' and raising C'. And the ratio of thermal rating to natural power, S_{th}/P_{nat}, sets the extent of the thermal plateau of the loadability curve of Section 5.6.

AC cables sit at $Z_w \approx 20 \ldots 40\,\Omega$, so their natural power exceeds the thermal rating and is never reached; as Section 5.3 sets out, the charging current binds first.

5.5.5 Light load: the Ferranti effect

On an open-ended line $\underline{I}_R = 0$ and the terminal voltage ratio follows from (25) as

$$\frac{\underline{U}_R}{\underline{U}_S} = \frac{1}{\cosh\left(\underline{\gamma} \cdot l\right)} \longrightarrow \frac{1}{\cos\left(\beta \cdot l\right)} \tag{34}$$

in the lossless limit, so the receiving end rises above the sending end. At $\beta \cdot l = 24.88°$ the open end reaches 1.102 p.u., drawing 251 MVAr of charging power. A 60 % shunt reactor brings it back to 0.984 p.u.; full compensation gives 0.918 p.u., Figure 11. The degree of compensation follows from the light-load voltage, but it also moves the natural power.

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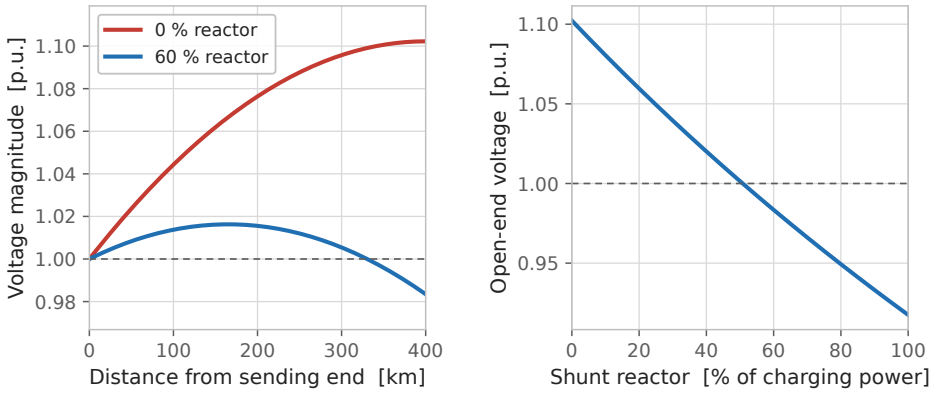


Figure 11: Open-end voltage rise against line length, and the effect of shunt compensation

5.5.6 Effect of compensation on the natural power

For uniformly distributed compensation the scaling is exact. Series compensation of degree k reduces the reactance and shunt compensation of degree m the capacitance:

$$X' \to X' \cdot (1-k) \;\Rightarrow\; \frac{P_{nat}}{\sqrt{1-k}}, \qquad C' \to C' \cdot (1-m) \;\Rightarrow\; P_{nat} \cdot \sqrt{1-m} \tag{35}$$

so $k = 60\,\%$ raises P_{nat} by a factor 1.581 and $m = 60\,\%$ lowers it by 0.632, Figure 12.

Real banks are lumped at substations rather than distributed, so these laws are a first-order guide only. A lumped 30 % series bank at $1.4 \cdot P_{nat}$ on the worked line reduces the transmission angle from $37.6°$ to $25.6°$ and turns the sending-end reactive flow from $+89$ to -30 MVAr. Section 5.8 treats the lumped case properly.

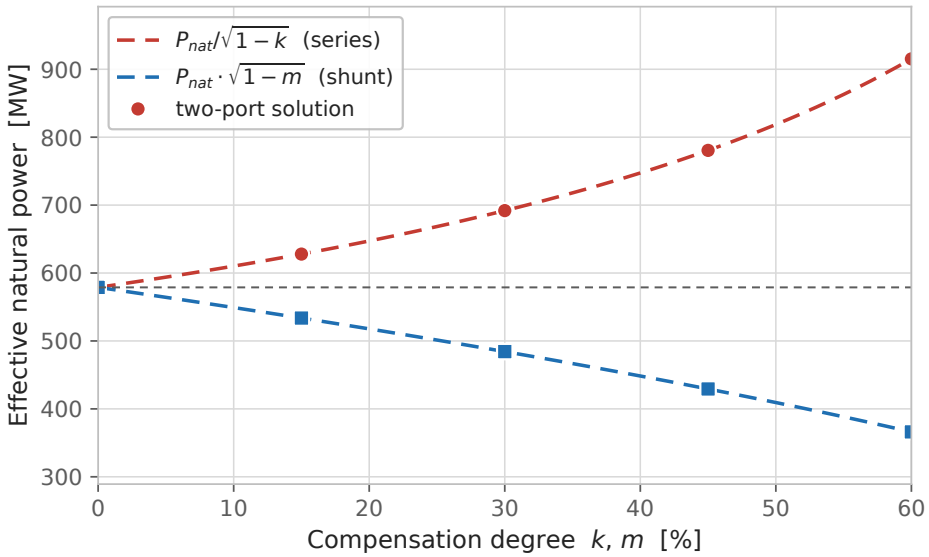


Figure 12: Effect of distributed series and shunt compensation on the natural power

5.6 Transmission Limits

The representation of Section 5.1 says what a line does, not what it may be asked to do. Three criteria bound the transfer, and which of them binds depends almost entirely on length.

5.6.1 The three criteria

Thermal. The conductor temperature sets a current, and with it a transfer that is independent of length:

$$P_{th} = \sqrt{3} \cdot U_n \cdot I_{th} \tag{36}$$

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The limiting mechanism is annealing of the aluminium and, more often, loss of ground clearance through sag. It is the only one of the three that is a property of the equipment rather than of the system, and it is weather- and time-dependent, which is what dynamic line rating exploits.

Voltage drop. The receiving-end voltage has to stay inside its band under load. Written on the two-port,

$$U_S = |\underline{A} \cdot \underline{U}_R + \underline{B} \cdot \underline{I}_R| \leq 1.0\,\text{p.u.} \quad \text{with} \quad U_R \geq 0.95\,\text{p.u.} \tag{37}$$

Beyond natural loading the line is a net consumer of reactive power, and the drop grows faster than the transfer. Shunt compensation at the receiving end raises this limit at the cost of a weaker voltage-stability margin.

Steady-state stability. The transmission angle must stay far enough below the crest of the power-angle curve that the system survives the loss of a parallel element. With a midpoint series bank of degree k the lossless form is

$$P_{stab} \approx \frac{U^2}{(1-k) \cdot X_L} \cdot \sin \delta_{max} \tag{38}$$

and a criterion of $\delta_{max} = 30°$ leaves a factor two on the lossless maximum. Section 5.7 treats this limit properly, including why the usable value is well below P_{max}.

The loadability of the line is the minimum of the three. Plotted against length it is the St. Clair curve.

5.6.2 Loadability of the reference line

Figure 13 evaluates the three criteria on the 500 kV reference line of Section 5.4 with the exact two-port relations, for two boundary conditions. In case (a) the load sits at the receiving end with no reactive support, which is the classical St. Clair configuration. In case (b) the line runs between two stiff systems with both terminals held at 1.0 p.u., which removes the voltage criterion entirely. The thermal limit is $I_{th} = 3 \times 1300$ A at 75 °C, giving $P_{th} = 3377$ MW against $P_{nat} = 939$ MW.

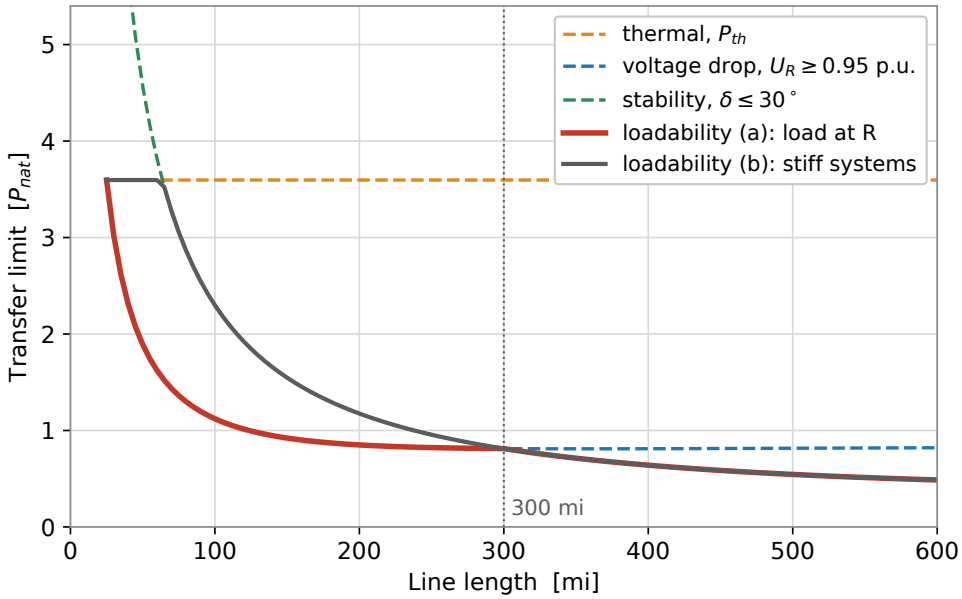


Figure 13: Loadability of the 500 kV reference line against length, cases (a) and (b), uncompensated

In case (a) the thermal limit binds only to about 25 mi, after which voltage drop takes over, and the loadability falls through $1 \cdot P_{nat}$ at 125 mi. In case (b) the thermal limit holds to about 60 mi and angular stability binds beyond it, with the crossing at 240 mi. At the 300 mi design length both cases give 763 MW, or $0.81 \cdot P_{nat}$ — the voltage and the angle criterion happen to coincide there, and the line cannot carry even its natural power without compensation.

The boundary condition is not a detail. Holding the receiving end at 1.0 p.u. moves the $1 \cdot P_{nat}$ crossing by 115 mi on this line. Any loadability figure has to be quoted with the reactive support it assumes.

5.6.3 Loadability of the 400 kV line

Figure 14 repeats the exercise on the 400 kV, 50 Hz line of Section 5.5, with criteria $I \le 2.0$ kA, $U_R \ge 0.95$ p.u. and $\vartheta \le 30°$. In case (a) the thermal limit holds to 80 km at $2.3 \ldots 2.4 \cdot P_{nat}$, voltage drop binds from 120 to 400 km, angular stability beyond 500 km, and the curve falls through $1 \cdot P_{nat}$ at 382 km. In case (b) the thermal limit holds to 160 km, stability binds beyond 200 km, and the crossing moves to 472 km.

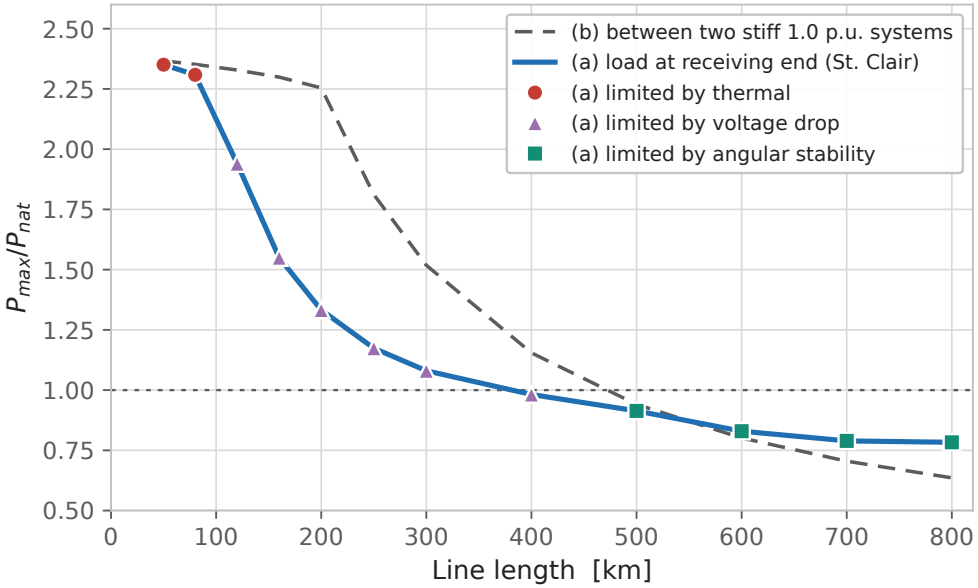


Figure 14: Loadability of the 400 kV line against length, with P_{nat} marked

The shape is the same on both lines: a thermal plateau of about $2 \ldots 2.5 \cdot P_{nat}$ over the first tens of kilometres, then a decay that crosses natural power somewhere between a quarter and a half of a thousand kilometres. That is the general statement the St. Clair curve makes.

5.6.4 Effect of series compensation

Figure 15 repeats case (b) for a midpoint series bank of degree k, and Table 17 gives the values at the 300 mi design length.

k	P_{stab}	P_U	case (b)	thermal to
	MW	MW	P_{nat}	mi
0	763	763	0.81	60
0.3	1027	826	1.09	85
0.5	1311	844	1.40	110
0.7	1708	796	1.82	145

Table 17: Effect of a midpoint series bank at 300 mi on the reference line

The bank acts on the stability limit: $k = 0.5$ lifts the 300 mi loadability of case (b) from 0.81 to $1.40 \cdot P_{nat}$. The voltage limit of case (a) barely moves, because the $I \cdot X_C$ rise across the bank is in quadrature with a unity-power-factor load. Series compensation therefore does not remove the need for reactive support at the receiving end; it addresses a different limit.

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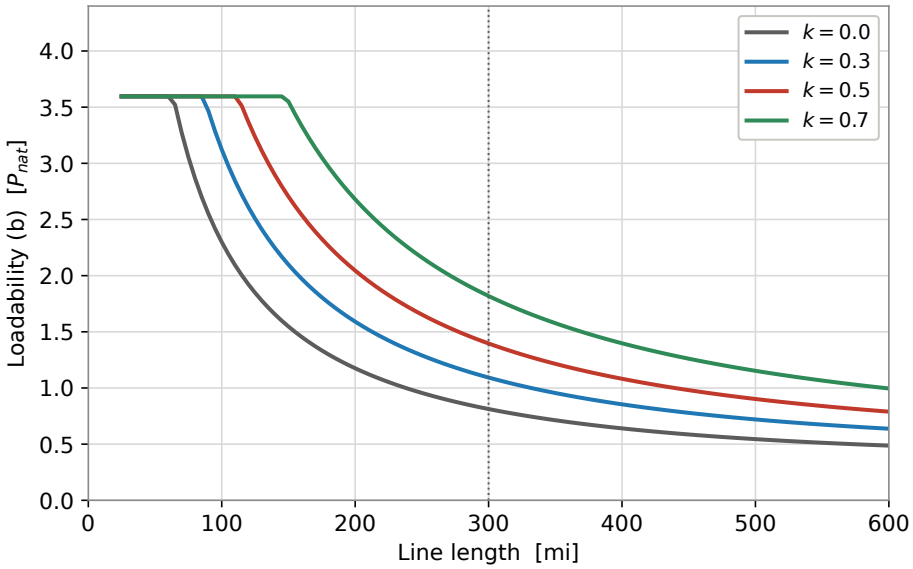


Figure 15: Case (b) loadability of the reference line with a midpoint series bank of degree k

5.6.5 Equipment limits

The conductor is not always the first thing to run out. Currents follow from the line-to-line voltage,

$$I = \frac{S}{\sqrt{3} \cdot U_{LL}} \tag{39}$$

and the switchgear in series with the line carries the same current. The largest continuous rating available in transmission circuit breakers is about 5000 A (Hitachi Energy, GE), so on a high-current scheme the breaker rating, not the conductor, can set the maximum current and force either a higher voltage or a second circuit.

5.7 Angular Stability

Section 5.6 introduced the angular criterion as one of three bounds. On any line long enough to matter it is the one that binds, and the margin it needs is larger than the criterion alone suggests.

5.7.1 Synchronising power

The meaningful quantity is not the crest of the power-angle curve but its slope, the *synchronising power coefficient*

$$K_s = \frac{\partial P}{\partial \delta} = \frac{U_S \cdot U_R}{X} \cdot \cos \delta \tag{40}$$

which is the stiffness of the electrical coupling between the machines at either end. Below the maximum, a rotor that advances delivers more power, decelerates and returns — negative feedback. Beyond it, advancing delivers less power and the machine accelerates further; $K_s = 0$ at the maximum is therefore the steady-state stability limit in the strict sense. Figure 16 shows both curves for the reference line.

With losses the maximum does not fall at $90°$ but at $\delta = \vartheta_B$, which is $81.0°$ on the reference line. At $\delta = 30°$ the line carries $0.81 \cdot P_{nat}$ with 79 % of the stiffness it has at $\delta = 0$.

Linearising the swing equation about an operating point gives

$$\frac{2 \cdot H}{\omega_0} \cdot \Delta\ddot{\delta} + K_D \cdot \Delta\dot{\delta} + K_s \cdot \Delta\delta = 0 \qquad \omega_n = \sqrt{\frac{K_s \cdot \omega_0}{2 \cdot H}} \tag{41}$$

with H the inertia constant and K_D the damping torque coefficient. As δ grows, K_s falls, the electromechanical mode slows and its damping ratio drops. The machine becomes slow and poorly damped before it loses synchronism, which is why the operating point is held below the maximum.

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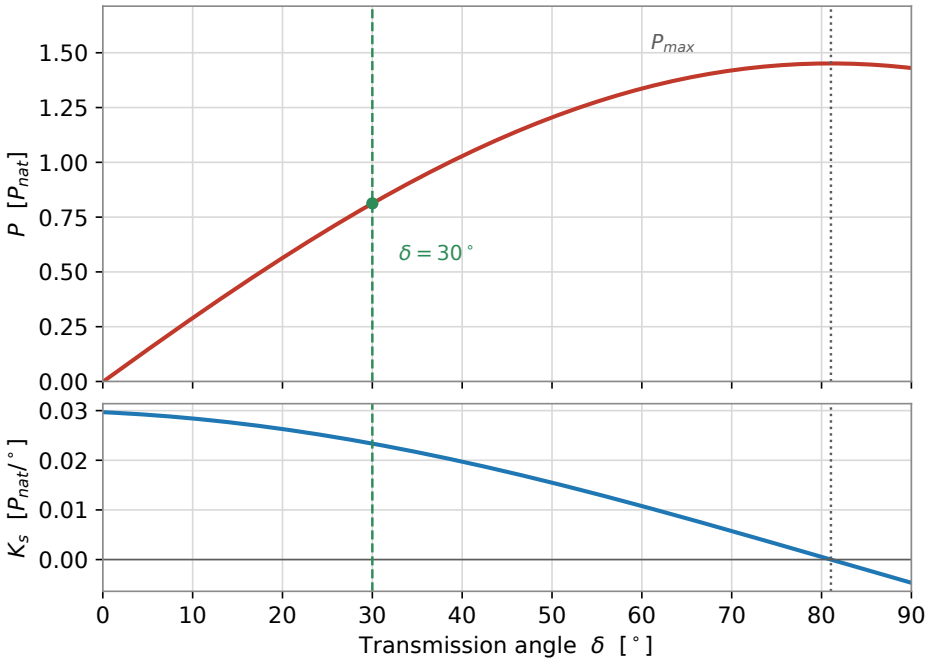


Figure 16: Power-angle relation and synchronising power coefficient of the 500 kV reference line at 300 mi, both terminals at 1.0 p.u.

5.7.2 Electrical length

Above about 250 km the lumped $\omega \cdot L' \cdot l$ overstates the reactance. The series element of the exact equivalent is $X' = Z_w \cdot \sin(\beta \cdot l)$, and substituting into the lossless maximum gives a clean result:

$$P_{max} = \frac{U^2}{Z_w \cdot \sin(\beta \cdot l)} = \frac{P_{nat}}{\sin(\beta \cdot l)} \tag{42}$$

At a quarter wavelength — 1234 km on the reference line at 60 Hz — the maximum transfer is exactly P_{nat}. That is the analytical spine beneath the St. Clair curve, and the reason capability decays with length rather than staying flat. Figure 17 compares the three against length.

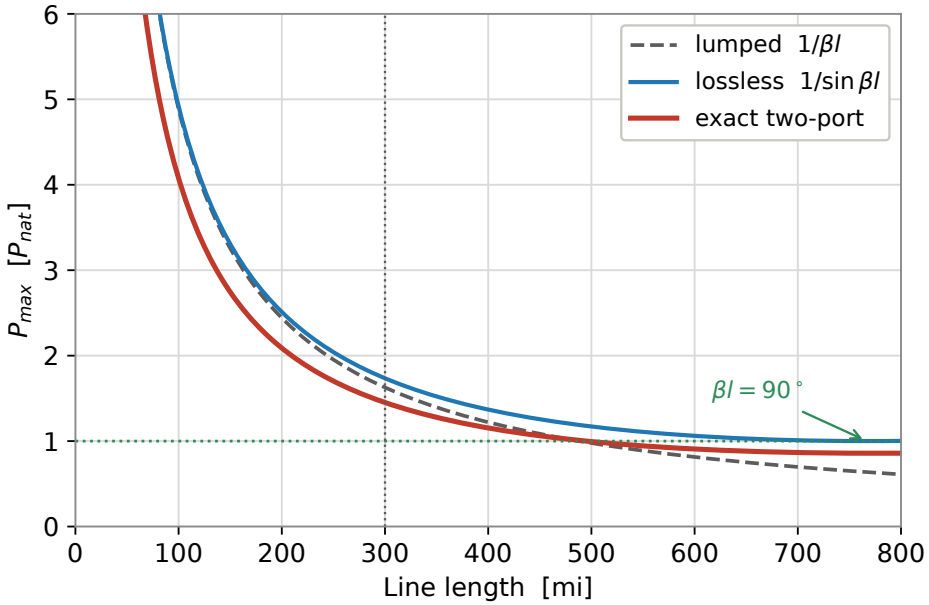


Figure 17: Maximum transfer against length: lumped, lossless distributed and exact two-port

At the 300 mi design length $\beta \cdot l = 35.2°$, the lumped reactance is $162.4\,\Omega$ against $X' = 153.4\,\Omega$, and

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$1/\sin(\beta \cdot l) = 1.73$ against 1.45 for the exact two-port; losses and charging account for the remaining gap.

5.7.3 Operating margin below the maximum

Two effects cut the usable transfer well below P_{max}, and they compound.

The first is the post-contingency network. Two parallel 300 mi circuits have $P_{max} = 2726$ MW intact; losing one doubles the reactance of the path and leaves 1363 MW on the survivor. Planning practice requires a margin of 30 to 35 % on the *post-contingency* network,

$$\frac{P_{max} - P}{P_{max}} \geq 0.30 \tag{43}$$

which on this pair allows 954 MW, or $1.02 \cdot P_{nat}$ for the two circuits together. A transfer of 1200 MW looks comfortable intact, at $\delta = 22.9°$, and sits only 12 % below the post-contingency maximum — not acceptable. For comparison, the thermal rating of the pair is 6755 MW, seven times the binding constraint. Figure 18 shows both power-angle curves.

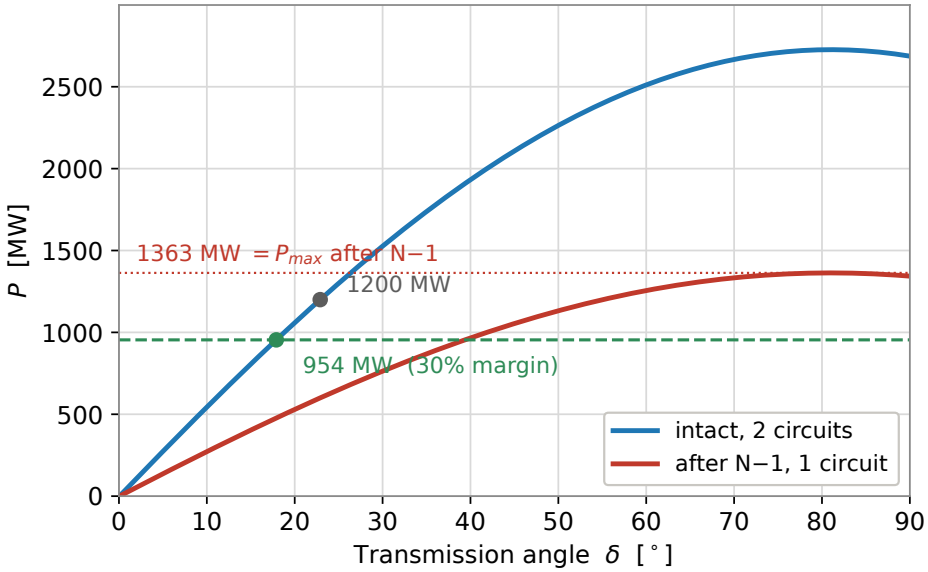


Figure 18: Power-angle curves of two parallel 300 mi circuits, intact and after the loss of one

5.7.4 Transient stability

The second effect is the fault case. During a fault the electrical output collapses while the mechanical input continues; the rotor must give that energy back before the maximum angle is reached. The equal-area criterion states the condition as

$$A_{acc} = \int_{\delta_0}^{\delta_{cc}} (P_m - P_{flt})\,\mathrm{d}\delta \;=\; A_{dec} = \int_{\delta_{cc}}^{\delta_{max}} (P_{pst} - P_m)\,\mathrm{d}\delta \tag{44}$$

with δ_{cc} the clearing angle. For a machine at $P_m = 0.6$ p.u. with maxima of 1.0 pre-fault, 0.3 during the fault and 0.7 after it, $\delta_0 = 36.9°$, $\delta_{cc} = 45.5°$ and $\delta_{max} = 121.0°$, Figure 19.

A higher pre-fault angle shrinks the decelerating area from both ends at once, so the critical clearing time falls steeply with loading. This is why two-cycle breakers and independent-pole tripping buy transfer capability, and why the transient limit usually binds before the steady-state one.

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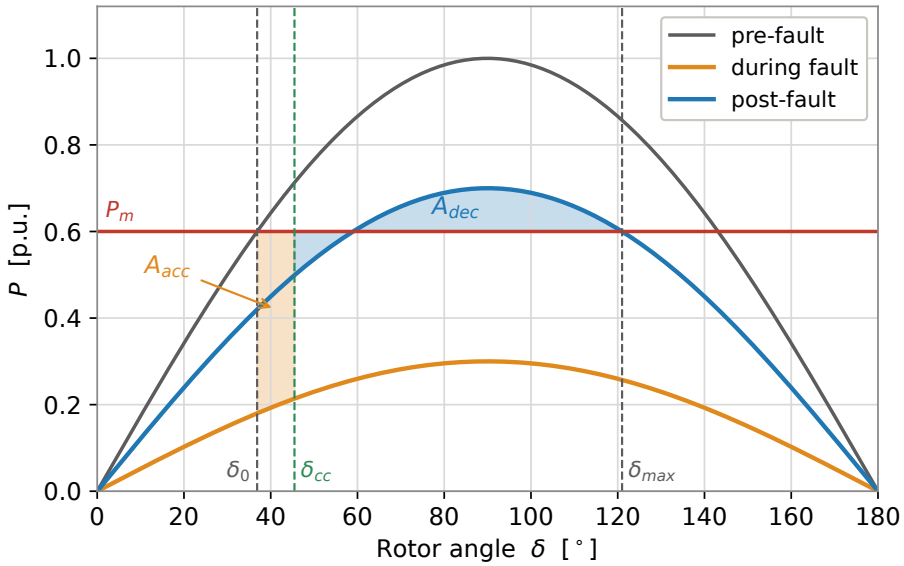


Figure 19: Equal-area criterion, generic machine per unit

5.7.5 Damping

In a multi-machine system the classical limit generalises to an eigenvalue criterion. The modes that constrain long transfers are the inter-area modes at 0.2 to 0.8 Hz; local plant modes at 0.8 to 2 Hz and torsional modes at 10 to 45 Hz. The practical limit is where the damping ratio of a critical mode falls below about 3 to 5 %, which can occur at angles below $40°$ and therefore below the two-machine maximum.

High-gain automatic voltage regulators, which permit a large angle, contribute *negative* damping torque at heavy loading. The power system stabiliser restores it. On a long corridor the available transfer is often released by tuning the stabiliser rather than by raising P_{max}.

5.7.6 Absence of a transmission angle on a DC link

A DC link has no transmission angle, no synchronising torque and no equal-area criterion. The converter control sets power as a commanded quantity, and length enters only through the conductor resistance, so the $1/\sin(\beta \cdot l)$ decay of (42) has no counterpart. An embedded DC link can be modulated to damp the inter-area modes of the surrounding AC system instead of loading them.

The constraint moves to the terminals: short-circuit ratio and commutation robustness for line-commutated converters, control interaction for voltage-sourced ones.

5.8 Series Compensation

5.8.1 Principle

A capacitor bank in series with the line cancels part of its inductive reactance. The degree of compensation is defined against the total series reactance,

$$k = \frac{X_C}{X_L}, \qquad X_L = \omega \cdot L' \cdot l, \qquad X_{eff} = (1 - k) \cdot X_L \tag{45}$$

so that the lossless transfer becomes

$$P = \frac{U^2}{(1 - k) \cdot X_L} \cdot \sin\delta \tag{46}$$

with $|U_S| = |U_R| = U$. The line behaves as though it were shorter: for a given transfer the angle falls and the stability margin of Section 5.7 rises.

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Two properties distinguish it from shunt compensation. The voltage rise $I \cdot X_C$ across the bank scales with the current, so the compensation is self-regulating: it appears when it is needed and not at light load. And it acts on the series branch, which is why it addresses the angular limit and barely moves the voltage limit of case (a) in Section 5.6.4.

The practical range is $k \approx 0.25 \ldots 0.70$. Above that, sub-synchronous resonance, distance-protection behaviour and bank cost dominate the choice.

In the two-port model the bank is cascaded as a series element at the point where it is installed,

$$\begin{bmatrix} 1 & -j \cdot X_C \\ 0 & 1 \end{bmatrix} \tag{47}$$

between the two half-lines for a midpoint bank, which is the arrangement used throughout this section.

5.8.2 Single-line arrangement

A series bank is not only capacitors. The platform sits at line potential and carries the full fault current until it is bypassed, so each phase carries a metal-oxide varistor across the bank to limit the capacitor voltage under fault current, a bypass breaker with a damping reactor to take over when the varistor reaches its energy limit, and disconnectors to isolate the platform. A bypass disconnector keeps the line in service with the bank out, at $k = 0$. Figure 20 shows the arrangement for one phase, with the bank at the midpoint of the line.

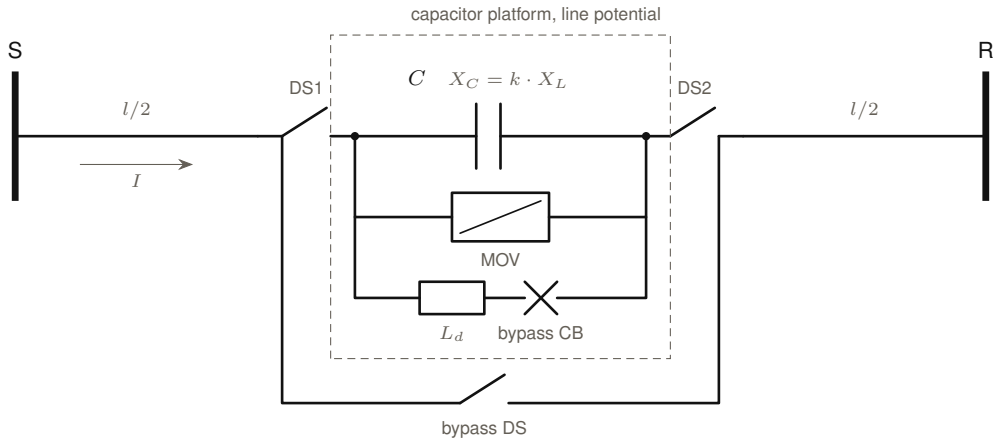


Figure 20: Single-line diagram of a midpoint series bank, one phase. Symbols to IEC 60617 [18]

The varistor limits the capacitor voltage under fault current and the bypass breaker with its damping reactor L_d takes over at the varistor energy limit. DS1 and DS2 isolate the platform; the bypass disconnector keeps the line in service with the bank out.

5.8.3 Effect on the power-angle curve

Figure 21 shows the power-angle curves of the 500 kV reference line for a midpoint bank at several degrees of compensation, and Table 18 the resulting maxima.

Uncompensated, 1500 MW — $1.6 \cdot P_{nat}$ — cannot be carried at all; at $k = 0.5$ it is carried at $36°$. The maximum also moves to smaller angles as k grows, which is the electrical shortening of the line expressed on the curve. The lossless form $U^2 / ((1 - k) \cdot X_L)$ would give 1540 MW uncompensated and 3080 MW at $k = 0.5$; at this length $R = 27.5\,\Omega$ and the line charging are not negligible, which is why the exact two-port is used.

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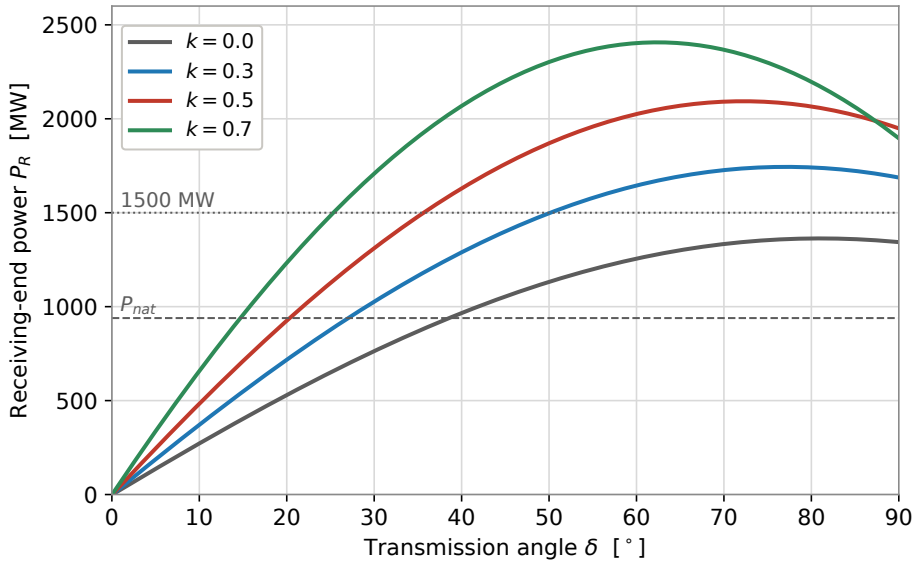


Figure 21: Power-angle curves of the reference line with a midpoint series bank, both terminals at 1.0 p.u.

k	X_C	P_{max}	δ **at 1500 MW**
	Ω	MW	°
0	0	1363	—
0.3	48.7	1744	50.1
0.5	81.2	2093	35.7
0.7	113.7	2407	25.3

Table 18: Midpoint bank on the 300 mi reference line, exact two-port

5.8.4 Voltage profile at heavy load

Delivering 900 MW at $\cos\varphi = 0.95$ lagging with the receiving end held at 1.0 p.u., the uncompensated line requires a sending-end voltage of 1.217 p.u., outside the usual 0.95 to 1.05 p.u. operating band. The midpoint bank raises the voltage by $I \cdot X_C$ in the direction of power flow, and the sending end falls to 1.157, 1.127 and 1.107 p.u. for $k = 0.3$, 0.5 and 0.7, Figure 22.

The receiving half of the line is unchanged, because the bank acts only on what is upstream of it. That is why end-of-line banks and split banks are also used where the voltage profile rather than the angle is the problem.

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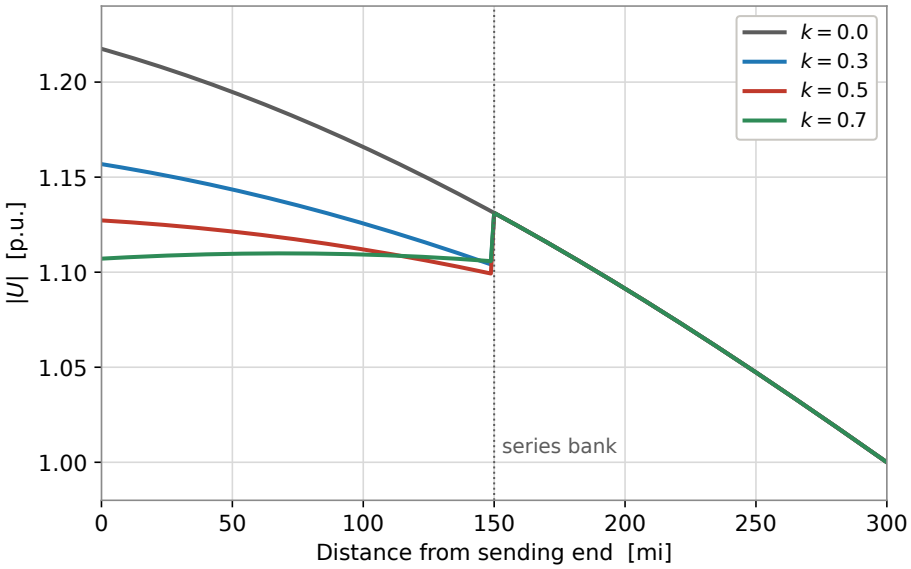


Figure 22: Voltage profile at heavy load with a midpoint series bank

5.8.5 Bank rating and protection

The bank is rated for the planned continuous line current, not for the conductor thermal limit:

$$U_C = I_{rated} \cdot X_C \qquad Q_C = 3 \cdot I_{rated}^2 \cdot X_C \tag{48}$$

Table 19 gives the resulting bank for the reference line at $k = 0.5$.

Quantity		Value
k		0.5
X_C	Ω	81.2
C per phase	µF	32.7
I_{rated}	A	2000
U_C per phase	kV	162
Q_C three-phase	MVAr	974
MOV level ($2.3 \cdot U_C$)	kV	373

Table 19: Midpoint bank rating on the reference line at $k = 0.5$

I_{rated} is taken as about $1.15\cdot$ the current at 1500 MW and 500 kV. The fault-current voltage is limited by the metal-oxide varistor at a protective level of $2.0 \ldots 2.5 \cdot U_C$, with the bypass breaker and damping circuit behind it. Emergency ratings — typically 30 min and 10 s — are specified separately from the continuous rating.

5.8.6 Sub-synchronous resonance

The bank forms a series resonance with the line and source reactances at

$$f_{er} = f_0 \cdot \sqrt{\frac{X_C}{X_L + X_S}} \tag{49}$$

Rotor currents see the complement $f_0 - f_{er}$, which has to stay clear of the torsional modes of the turbine-generator shaft — 10 to 45 Hz for large steam units. Figure 23 screens the reference line against the degree of compensation.

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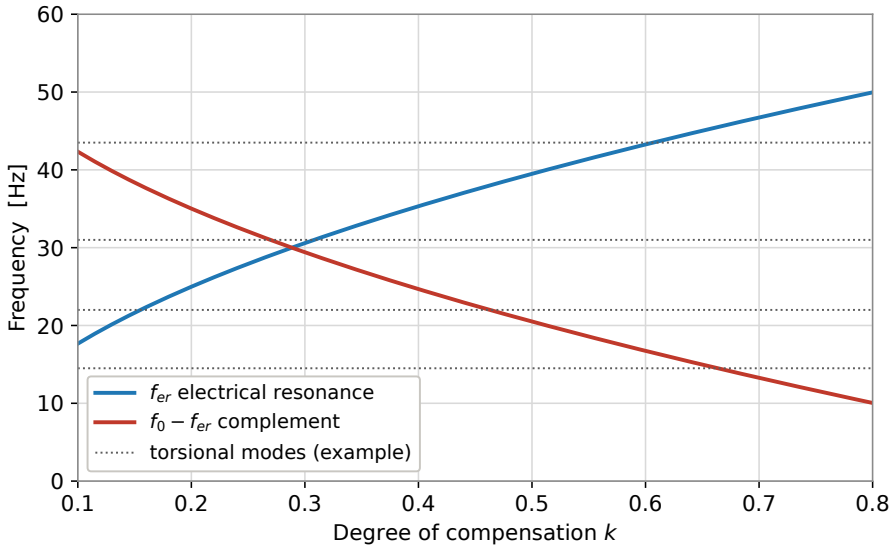


Figure 23: Sub-synchronous resonance screening: electrical resonance and its complement against the degree of compensation

With $X_S = 25\,\Omega$, which is 15 % on 1500 MVA, a compensation of $k = 0.5$ gives $f_{er} = 39.5$ Hz and a complement of 20.5 Hz — within 2 Hz of an illustrative 22 Hz torsional mode. A hit at screening level calls for a detailed study or a different k; type-3 wind generation adds sub-synchronous control interaction to the same screening. The torsional modes used at this stage are illustrative, and real values come from the generator manufacturer.

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6 DC Fault Ride-Through

A fault on the DC side of a VSC HVDC link interrupts the power transfer. On a link whose DC circuit is at least partly an overhead line, most of these faults are temporary flashovers — lightning, pollution, vegetation — and the link is expected to clear the fault and restore transmission on its own, as an AC line does by auto-reclosure. This chapter describes that behaviour for a half-bridge MMC in a symmetrical monopole, on the record of the Pugalur–Thrissur link in India. The supplier's description of the design [24] and the operator's evaluation of five years of fault records [25] are the sources; every number in this chapter is theirs.

The term used throughout is the one of the project specification: a *DC line fault recovery sequence*, a defined series of restart attempts after a fault on the DC line. The expression *DC fault ride-through* in the chapter title names the same behaviour from the point of view of the AC network.

6.1 Reference project

The Pugalur–Thrissur link connects Pugalur (Tamil Nadu) and North Thrissur (Kerala) with two independent symmetrical monopoles of 1000 MW each, at ±320 kV, bidirectional, with half-bridge MMC converters and a reactive power capability of ±1100 Mvar [25]. The DC circuit is a hybrid of 150 km overhead line and 28 km XLPE cable; a transition station 28 km from Thrissur joins the two [25] (the project map in [24] gives 143 km and 32 km). At Pugalur the link shares the 400 kV AC bus with the ±800 kV, 6000 MW LCC link from Raigarh. It was commissioned in 2021. Table 20 lists the system parameters the recovery performance was specified against.

Parameter		Pugalur	Thrissur
AC voltage, max./min.	kV	440 / 360	440 / 360
AC frequency, max./min.	Hz	52.5 / 47.5	52.5 / 47.5
Negative-sequence voltage	%	1.5	1.5
Short-circuit power, max./min.	MVA	36 400 / 390	36 400 / 390
DC voltage, pole to pole	kV	640	
DC current	A	1563	
Sub-module type		half-bridge	

Table 20: System parameters of the Pugalur–Thrissur VSC HVDC link [25]

6.2 Fault behaviour of the half-bridge converter

A blocked half-bridge MMC is an uncontrolled diode rectifier between the AC network and the DC circuit. Blocking the converter therefore neither interrupts the fault current nor de-energises the DC circuit: the AC network keeps feeding the fault through the diodes until the AC circuit breakers of both stations open [24]. Opening the AC circuit breakers is thus a mandatory part of every DC fault clearing on this converter type, and the mechanical switchgear of Section 6.4 sets the pace of the recovery.

The symmetrical monopole is the DC equivalent of an ungrounded AC network. Its ground reference is a high-impedance star-point unit, which grades the two pole voltages against ground, so the two fault types behave differently [24]:

- **Pole-to-ground fault.** The high grounding impedance limits the fault current to the discharge of the faulted pole and the charging of the healthy one. The healthy pole, however, rises towards the full pole-to-pole voltage: the records of [25] show up to 600 kV on a ±320 kV link, limited by the arresters. The overvoltage persists after the AC circuit breakers open, because the cable section stores the charge, and has to be removed by a discharge unit before the restart.

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- **Pole-to-pole fault.** There is no overvoltage, but a fault current fed by the diode rectifiers of both stations. After the AC circuit breakers open, this current freewheels through the converter diodes, driven by the energy in the converter reactors and the DC circuit, and decays with the L/R time constant of the fault loop. With no fault impedance the decay takes up to 500 ms [24]. The deionisation time starts only when the current has reached zero.

The fault detection and the blocking are independent of the fault location: the IGBTs have to be switched off within microseconds, before the current exceeds their turn-off capability, which rules out any selective decision at that stage [24]. Selectivity — in which zone the fault lies and whether a restart is permitted — is decided afterwards, during the fault clearing, Section 6.7.

6.3 Specified recovery performance

The technical specification of the project defines the recovery as a sequence of up to three restart attempts, each with its own target state and time limit [24], [25]:

1. **First restart** at the pre-fault direct voltage and power transfer, within 1.5 s of a pole-to-ground fault and 1.8 s of a pole-to-pole fault, both including fault clearing and cable discharge.
2. **Second restart** at a reduced direct voltage, with the power curtailed accordingly. Its duration is governed by the on-load tap-changer and the recharging of the switchgear drives, and is specified only as slow, above 10 s.
3. **Changeover to STATCOM operation** if the second restart fails: the DC line is de-energised permanently and both converter stations reconfigure to reactive power support of the AC network, again above 10 s.

The STATCOM requirement comes from the AC network: the loss of the transmission changes the load flows and raises the need for reactive power, and the converter supplies it without additional FACTS equipment. The same requirement applies after a permanent fault in the DC cable or in one of the converter stations [24].

A fault within 15 min of the previous one counts as a consecutive fault of the same sequence, and only the remaining attempts are made; after 15 min without a fault the sequence is complete and the next fault opens a new one with all three attempts. Three such sequences have to be possible within 45 min [24], [25]. As the project has two parallel links, the specification also requires the active power of the faulted link to be transferred to the healthy one, as far as its rating allows, within 80 ms [24]. Table 21 collects the times.

Step	Specified time
First restart, pole-to-ground fault (incl. fault clearing and cable discharge)	< 1.5 s
First restart, pole-to-pole fault	< 1.8 s
Second restart at reduced direct voltage	> 10 s (tap-changer, drive recharge)
Changeover to STATCOM operation	> 10 s
Power compensation by the parallel link	< 80 ms
Consecutive fault	within 15 min of the previous one
Recovery sequences to be withstood	3 within 45 min

Table 21: Recovery performance specified for the Pugalur–Thrissur link [24], [25]

6.4 Switchgear of the recovery sequence

Figure 24 shows the single-line diagram of one converter station with the switchgear the recovery sequence operates [24], [25]. Four devices take part:

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- **AC circuit breaker.** Clears the fault by separating the diode rectifier from the AC network, and re-energises the interface transformer and the converter at the restart.
- **Insertion resistor and bypass switch.** The insertion resistor is in series with the network terminals of the interface transformer and is bypassed in normal operation. For the restart the bypass switch is opened before the circuit breaker closes, so the DC circuit is charged through the resistor and the diodes. The resistor limits the inrush of the diodes and, if the fault persists, the uncontrolled fault current. Once the charging current has decayed the bypass switch closes and the converter is deblocked.
- **Discharge unit.** One per pole: a fast-acting grounding switch in series with a damping resistor. It removes the trapped charge of the healthy pole after a pole-to-ground fault, and is closed only for that purpose. The resistance is chosen well above the surge impedance of the DC circuit so that the discharge does not launch travelling waves of its own [24]. At Pugalur–Thrissur the discharge is led by the Thrissur station, the Pugalur unit being the backup [25].
- **DC disconnect switches.** Slow devices that separate a converter station from the DC circuit for STATCOM operation. Their operating time, 15 s and more, is one of the two terms that set the changeover time of Section 6.8.

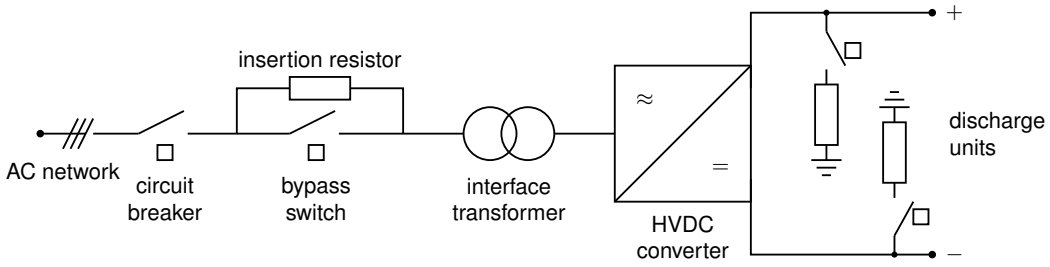


Figure 24: Switchgear of one converter station taking part in the DC line fault recovery, after [24]: AC circuit breaker, insertion resistor with bypass switch, interface transformer, converter and one discharge unit per pole

6.5 Recovery sequence

The first restart after a pole-to-ground fault on the overhead line runs in eight steps, coordinated between the two converter stations and the transition station over the HVDC telecontrol; without telecontrol the sequence cannot start [25].

1. **Fault isolation.** The converters block on fault detection and the AC circuit breakers of both stations open. Until they open, the fault current flows through the diode path.
2. **Fault current decay.** With the breakers open the fault current decays with the L/R time constant of the loop, set mainly by the star-point units. The overvoltage of the healthy pole, limited by the arresters, discharges slightly.
3. **Discharge and deionisation.** The discharge unit of the lead station closes and removes the charge of the healthy pole; the bypass switch of the insertion resistor opens. The arc at the fault location deionises.
4. **Condition verification.** Both stations check breaker and bypass-switch positions, arm, DC and star-point currents and DC voltages against their thresholds.
5. **Tap-changer adjustment.** Only for a restart at reduced voltage: the tap-changers move to the start position given by the sequence counter.
6. **Restart initiation.** The restart release closes the AC circuit breaker of the rectifier station first, then of the inverter station.
7. **Passive charging.** The DC circuit charges through the insertion resistor and the diodes. The build-up of the voltages and the decay of the charging current confirm the integrity of the DC circuit before

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the resistor is bypassed or the converter is deblocked [24]. Since the sub-module capacitors are still charged, this step is shorter than a cold start.

8. **Converter deblocking.** The converters deblock and ramp to the reference direct voltage. The rectifier station restores its reactive power first; the inverter station then restores active and reactive power to the pre-fault values.

Figure 25 gives the durations of the four phases as the operator reports them from the fault records [25]. The recovery from a pole-to-pole fault omits the discharge unit, since there is no trapped charge, but has to wait for the freewheeling current to reach zero before the deionisation time starts, Section 6.6.2.

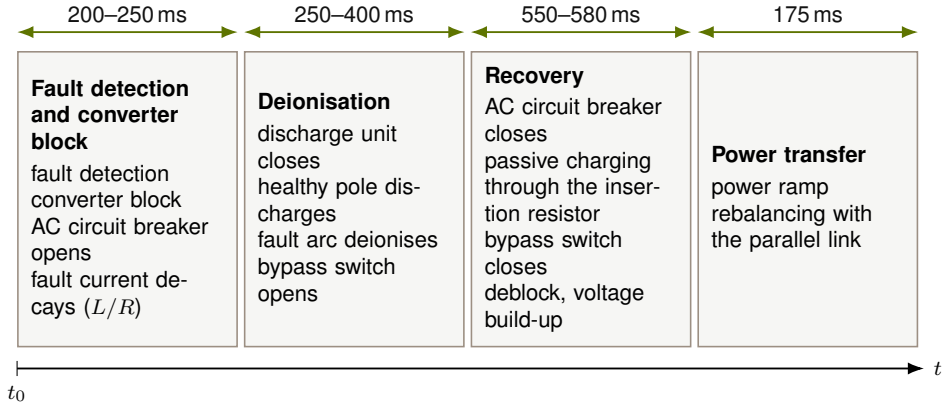


Figure 25: Phases and durations of the first restart after a pole-to-ground fault on the DC overhead line, from the fault records of the Pugalur–Thrissur link [25]

6.6 Recorded recoveries

6.6.1 Pole-to-ground fault

The commissioning record in [24] shows a first restart with both links at about 50 % active and reactive load and the fault on the positive pole. The converters of the faulted link block at the fault instant t_0 and the healthy link takes over the power within the specified time. The negative pole is discharged at $t_0 + 180$ ms, the circuit breaker of station A closes and passive charging starts at $t_0 + 520$ ms, the bypass switch closes at $t_0 + 700$ ms with the deblocking of station A, and station B deblocks at $t_0 + 915$ ms and restores its reactive power. The active power sharing between the two links is then restored with a long time constant, so that the AC network sees no second disturbance; this slow return applies only to the share the healthy link took over, and a share it could not take over is restored immediately after deblocking [24]. Figure 26 is that record: the pole voltages U_{dHP}, U_{dHN}, the pole currents I_{dHP}, I_{dHN}, the active power of the two links and their sum, and the reactive power of the two links at each station. The negative pole holds about -580 kV until the discharge unit closes, and the active power of link 2 (red) steps up to carry the total while link 1 (blue) recovers.

Table 22 lists the three pole-to-ground faults the operator evaluated from the transient fault recorder at Pugalur after five years of service, together with the dynamic performance study case the control design was validated against [25]. The three field events share the same sequence: the healthy pole rises to 600 kV and is discharged until $t_0 + 250$ ms, the deionisation takes about 280 ms, zone identification and the reconfiguration request fall into these two intervals, the circuit breaker recloses with the insertion resistor in circuit for about 200 ms, and the converter deblocks between $t_0 + 860$ ms and $t_0 + 930$ ms. Power is restored to the pre-fault level within 2 s in each case, and the parallel link carries the power in the meantime. The simulated case at rated power deblocks within 800 ms. All recorded first restarts succeeded, and all lie within the specified 1.5 s [25].

Figure 27 is the transient fault record of the first event, at 400 MW on link 1. The positive pole is the healthy one here and rises to 600 kV at the fault; the cursors mark 0.722 s from the fault to the reclosing of the circuit breaker, 0.193 s of charging through the insertion resistor, and 0.932 s to deblocking. The binary channels

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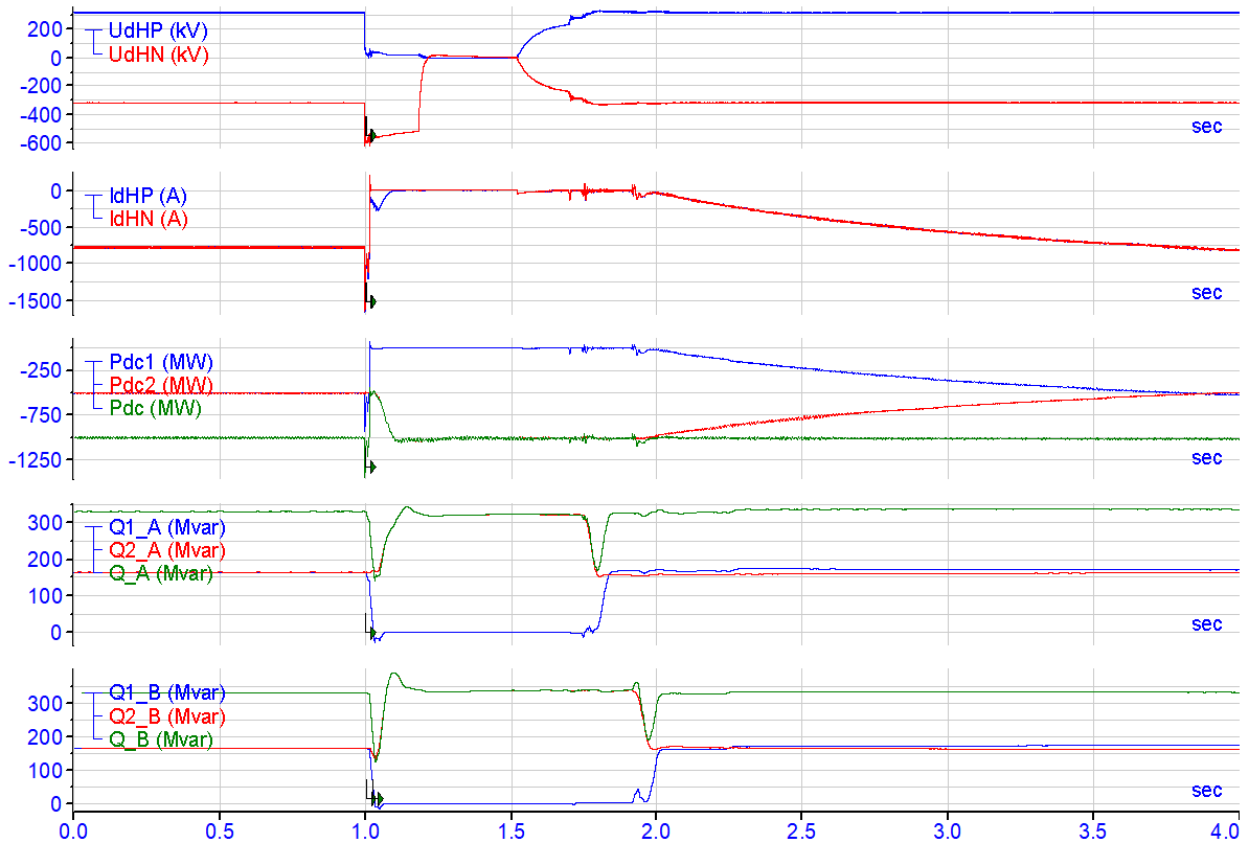


Figure 26: Commissioning record of the first restart after a single pole-to-ground fault on the positive pole, both links at about 50 % load; from top: pole voltages, pole currents, active power of link 1, link 2 and total, reactive power of both links at station A and at station B. Reproduced from [24], Figure 2

below the analogue traces show the blocking, the zone identification (*ZONE = DC OHL*), the reconfiguration request and the restart release, and the positions of the circuit breaker, the bypass switch and the discharge switch. Figure 28 is the dynamic performance study case at rated power the controls were validated against, with the same sequence and a deblocking within 800 ms.

Case	P	Q	Current	Faulted pole	Deblock	Power
	MW	Mvar	A		ms	s
Link 1, field	400	-150	625	negative	930	2
Link 2, field	250	0	390	positive	890	2
Link 2, field	400	0	625	negative	860	2
Simulation, rated	1000	$+330$	1563	positive	< 800	≈ 2

Table 22: Recorded and simulated first restarts after a pole-to-ground fault on the Pugalur–Thrissur link, reverse power flow Thrissur to Pugalur; deblocking time after t_0 and time to pre-fault power [25]

6.6.2 Pole-to-pole fault

The commissioning record of a pole-to-pole fault in [24] shows the fault current fed through the diode rectifiers after blocking, with the corresponding reactive power drawn from the AC network as for a three-phase fault with residual voltage. After the circuit breakers open the current decays and reaches zero in both stations at $t_0 + 145$ ms. The circuit breakers reclose at $t_0 + 850$ ms and passive charging begins; station A deblocks at $t_0 + 1.03$ s and station B at $t_0 + 1.25$ s, with the same restart procedure as after a pole-to-ground fault. The sequence contains no discharge step, but it has to allow for the current decay of up to 500 ms, Section 6.2, and for the deionisation time after it before the circuit breakers reclose [24]. Figure 29 is the

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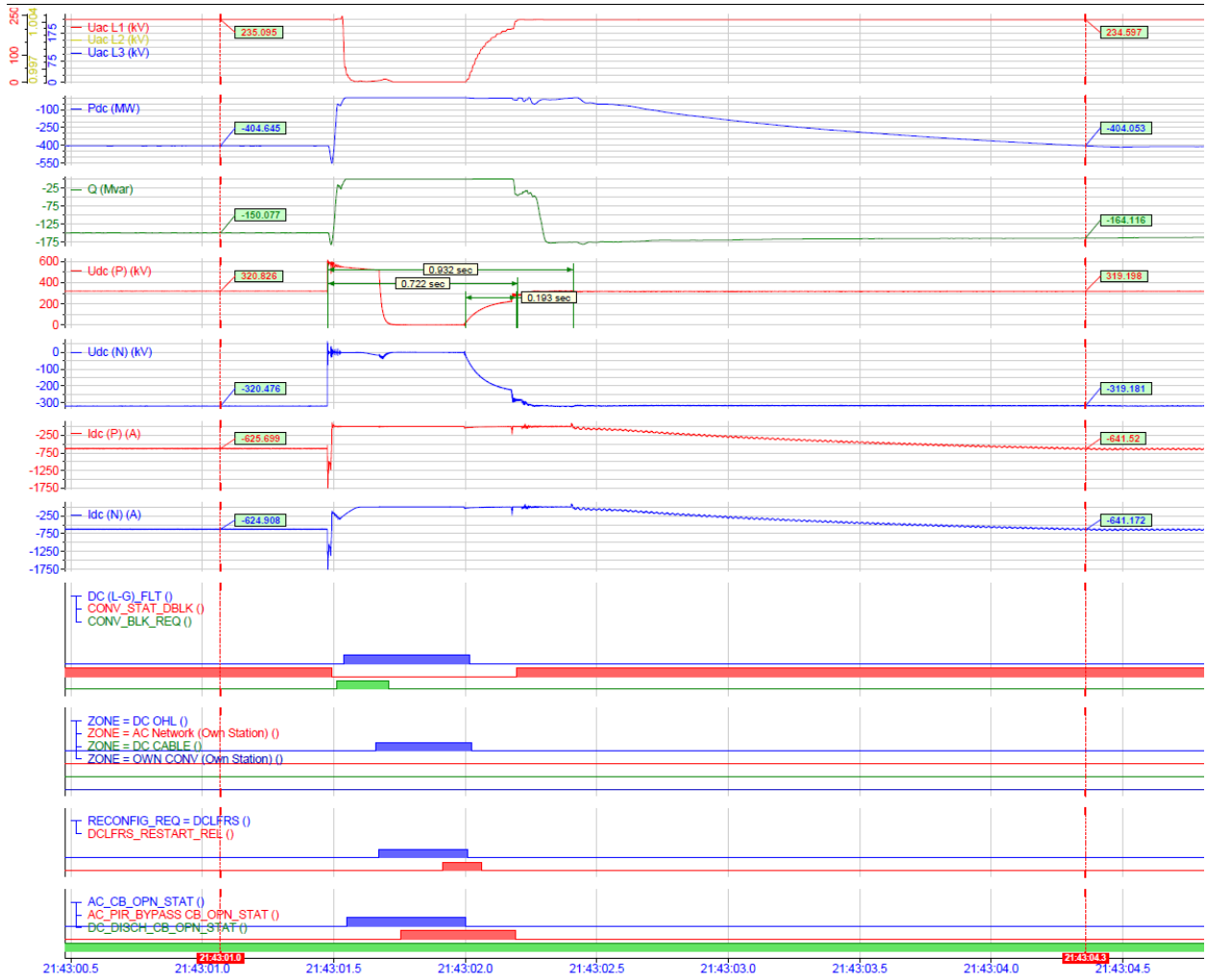


Figure 27: Transient fault record at Pugalur of the pole-to-ground fault on the negative pole of link 1 at 400 MW; from top: AC phase-to-ground voltages at the interface transformer, active power, reactive power, positive and negative pole voltages, positive and negative pole currents, and the binary channels of the protection and the switchgear. Reproduced from [25], Figure 7

record: both pole voltages collapse to zero without overvoltage, the pole currents peak at about -10 kA before the breakers open, and the reactive power of both stations shows the three-phase fault seen from the AC side.

6.6.3 Recovery at reduced direct voltage

The second restart recovers to a reduced direct voltage. A half-bridge MMC has a narrow direct-voltage range: to avoid over-modulation, the direct voltage to ground has to stay above the peak of the AC voltage the converter produces at its terminals, so the AC-side voltage has to be lowered with the direct voltage. The ratio of the interface transformer is changed with its on-load tap-changer, before the circuit breaker recloses, since the link has to recover to the reduced voltage immediately after deblocking. Depending on the number of taps this takes up to 1 min [24]. The operator preselects the reduced voltage between ± 256 kV and ± 305 kV; a link already running below the preselected level is kept from being raised to it by the recovery [24]. The commissioning record of a second restart from ± 320 kV to ± 256 kV, with the other link out of service, shows the power restored about 60 s after the fault [24], Figure 30; the time axis of that record spans 70 s, against 4 s for the first restarts of Figures 26 and 29.

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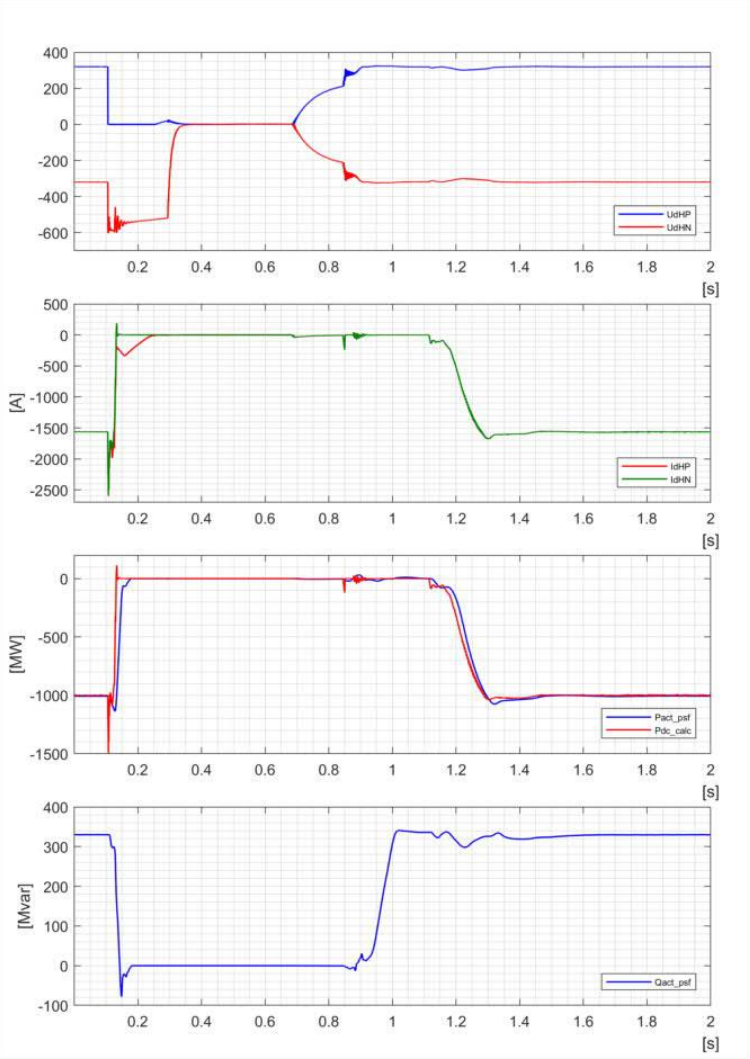


Figure 28: Dynamic performance study of a pole-to-ground fault on the positive pole at 1000 MW, $+330$ Mvar; from top: pole voltages, pole currents, active power and reactive power. Reproduced from [25], Figure 10

6.7 Fault zones and selectivity

A restart is permitted only for a fault on the overhead line. The protection of each link is therefore divided into the fault zones AC network, converter station, DC overhead line and DC cable at each end, Figure 31, with the HVDC link zone spanning both converters and the DC circuit [24], [25]. Voltage and current measurements at the zone boundaries — current transformers at the transition station separate the overhead line from the cable — produce fault zone identifiers that state on which side of the measuring point the fault lies. A logic combines the identifiers from all locations while the fault clearing is in progress, so that the zone is known before the recovery sequence is released [24]. On the Pugalur records the zone identification and the reconfiguration request fall into the discharge phase, before $t_0 + 250$ ms [25].

Two identifier algorithms cover the range of fault impedances an overhead line produces [24]. Faults of low impedance are detected by a wave-front logic that evaluates the polarity of the instantaneous changes of the direct current. Faults of high impedance are detected by a residual-current logic acting on the discharge currents of the DC cable; it covers fault impedances up to 5 kΩ.

6.8 Reconfiguration to STATCOM operation

The outcome of each attempt is classified by the passive charging and the following 15 min [24]. A fault detected anywhere in the link during a restart attempt makes the fault *permanent*, and no further attempt

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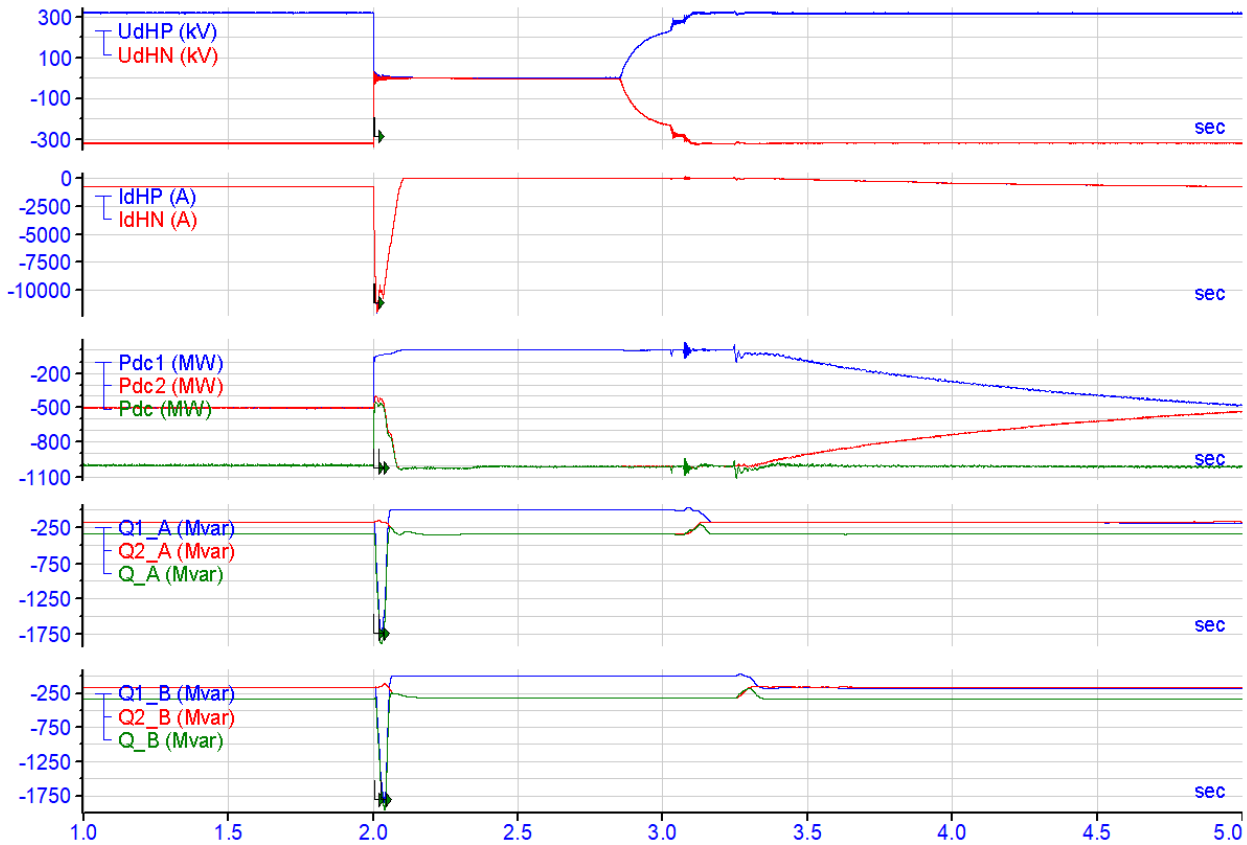


Figure 29: Commissioning record of the first restart after a pole-to-pole fault on link 1; from top: pole voltages, pole currents, active power of link 1, link 2 and total, reactive power at station A and at station B. Reproduced from [24], Figure 4

follows: on a persisting pole-to-ground fault the direct voltage builds up in one pole only after the circuit breaker closes, the current stays limited by the insertion resistor, the protection detects the condition, the start-up is aborted and the healthy pole is discharged, Figure 32. An attempt is *successful* when the power is restored; a fault after that point is *consecutive* and starts the next attempt. When 15 min pass without a fault the sequence is complete.

A sequence ends unsuccessfully either by a permanent fault or by a third consecutive fault. In both cases both stations change over to STATCOM operation after the fault clearing: the DC disconnect switches separate each converter from the DC circuit and each station restarts independently [24]. The changeover time is the sum of the operating time of the slow DC disconnect switches and, when the link was running at rated voltage, of the tap-changer movement to the STATCOM position; it ranges from 15 s to about 1 min. In the commissioning record of [24] the link was already at reduced voltage, the disconnect switches were the only term, and both stations restored reactive power within about 14 s, Figure 33. The converters are therefore unavailable for the transient voltage support in the first seconds after the loss of transmission and available for the steady-state support afterwards.

The same changeover is started directly, without any restart attempt, when the fault zone is the DC cable [24], [25]. A fault in one converter station prohibits the changeover for that station, and only the other one reconfigures to STATCOM operation [24].

6.9 Ratings affected by the recovery

The short-circuit level of the AC networks acts in opposite directions on the two fault types [24]: the pole-to-pole fault current rises with the short-circuit level, whereas the pole-to-ground overvoltage rises as the short-circuit level falls, because the load rejection of the faulted link raises the AC voltage more in a weak network. The pre-fault active and reactive power have no comparable influence on the fault behaviour beyond this overvoltage effect. The commissioning tests of [24] were run at short-circuit levels above

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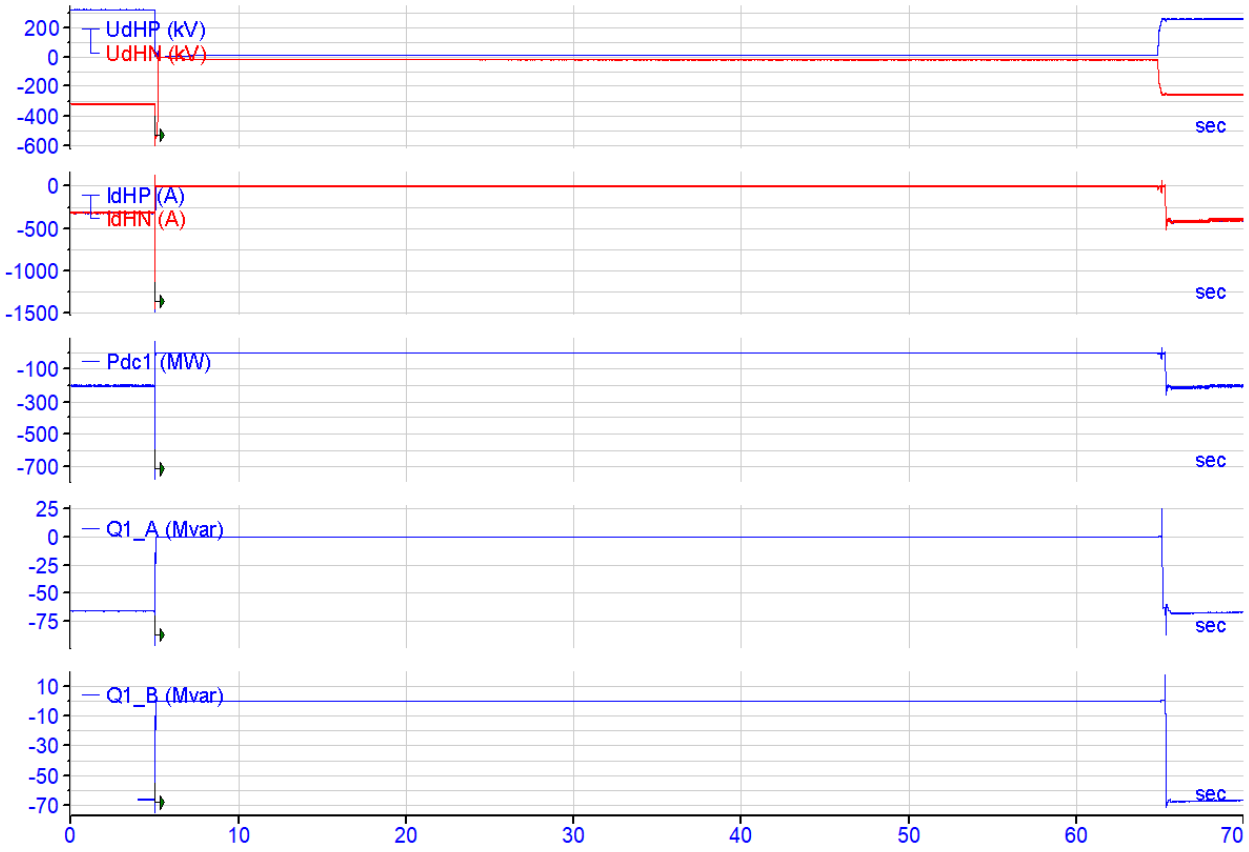


Figure 30: Commissioning record of a second restart to the reduced direct voltage of ±256 kV after a pole-to-ground fault, other link out of service; from top: pole voltages, pole currents, active power of link 1, reactive power at station A and at station B. Reproduced from [24], Figure 5

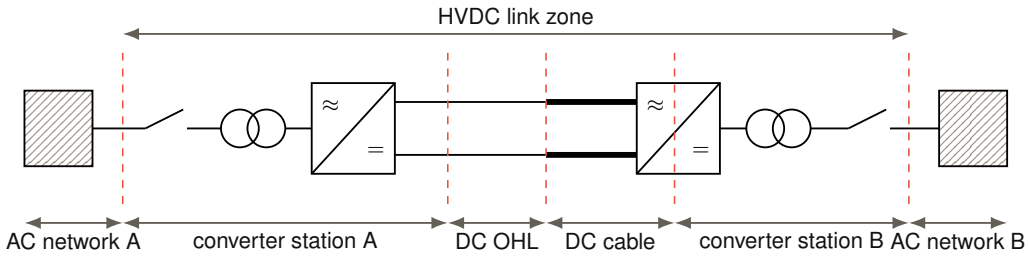


Figure 31: Fault zones of one link of the Pugalur–Thrissur project, after [24], [25]; the thick section is the DC cable between the transition station and station B

10 GVA and at a pre-fault power the healthy link could compensate in full. The equipment ratings and the protection settings were fixed against a defined relation of network overvoltage to short-circuit level, so that the extreme network conditions are covered without over-rating [24].

The cable section raises the fault rate of the DC circuit, since the overhead line is exposed, while it is the cable that has to carry the fault currents and the overvoltages [24]. The currents through the cable screens and the grounding system, the overvoltages across the main insulation and the polarity reversals enter the cable rating. Measurements at the Thrissur cable termination for faults on the overhead line near Pugalur show the $\mathrm{d}u/\mathrm{d}t$ of the initial voltage collapse reduced by the cable impedance, a well damped response to the pole-to-ground fault, and a first voltage reversal of about 40 % for the pole-to-pole fault, Figure 34; steep fronts lie outside the bandwidth of the installed recorders and were covered in the design studies [24].

Each fault and recovery is a thermal event. The devices with long thermal time constants — arresters, insertion and discharge resistors, and the star-point unit — are rated for the specified number of recoveries, three sequences within 45 min, and their thermal state is monitored by the protection [24]. The star-point unit also sets the steady-state grading of the two pole voltages: the unsymmetrical leakage of a DC overhead

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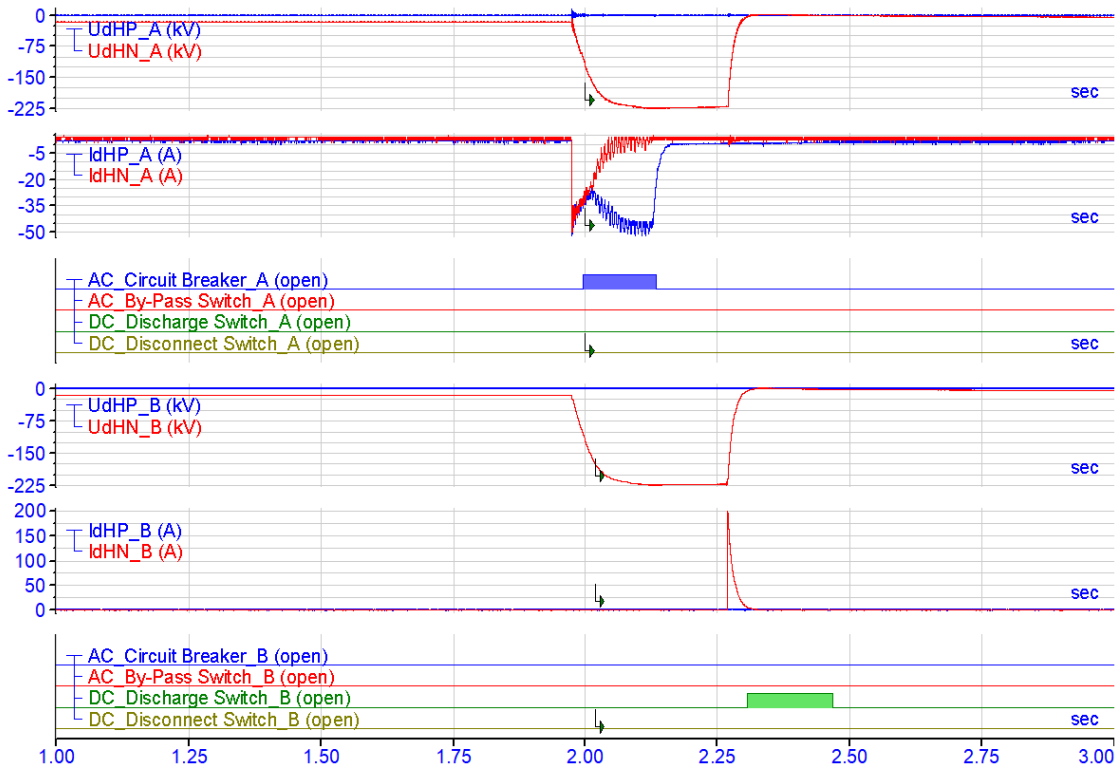


Figure 32: Commissioning record of a restart attempt on a permanent pole-to-ground fault on the positive pole: after the circuit breaker of station A closes, the voltage builds up in the negative pole only, the currents stay limited by the insertion resistor, the attempt is aborted and the discharge switch of station B removes the healthy-pole voltage; for each station the pole voltages, pole currents and the positions of circuit breaker, bypass switch, discharge switch and DC disconnect switch. Reproduced from [24], Figure 8

line, from insulator and corona currents, was specified as a maximum grading current unbalance for this project, and for considerably longer overhead sections the supplier states that the present grading devices may not remain an economic solution [24].

6.10 Design alternatives

Four alternatives were evaluated at the offer stage of the project and rejected on cost [24].

- **Full-bridge sub-modules.** A full-bridge MMC is an open circuit when blocked, so the AC circuit breaker need not open and the AC network is less disturbed, and it operates at a modulation index above one, so it recovers to a reduced direct voltage without the tap-changer. Both remove mechanical operations from the sequence and shorten the recoveries and changeovers. The alternative was set aside for its higher investment and losses, and because few bidders could offer it.
- **DC circuit breaker.** Clears the fault in the DC circuit and may keep the converters in service. Precautions for the DC overvoltage and the cable inrush remain and enter the achievable recovery time. Rejected for investment cost and the lack of competing bidders.
- **Fixed no-load voltage.** A converter-side no-load voltage chosen for the minimum direct voltage at the maximum modulation index allows recovery to any direct voltage within the same 1.5 s or 1.8 s, without tap-changer movement. The converter then operates at a higher AC current, with operating losses about 20 % higher. The project chose the losses over the recovery time.
- **Fast disconnect switches.** Would reduce the changeover to STATCOM operation to the few seconds of the fault clearing, but only when the link was already at reduced voltage, since otherwise the tap-changer sets the time; their price is 5 to 10 times that of slow disconnects, and the probability of a fast changeover was judged low.

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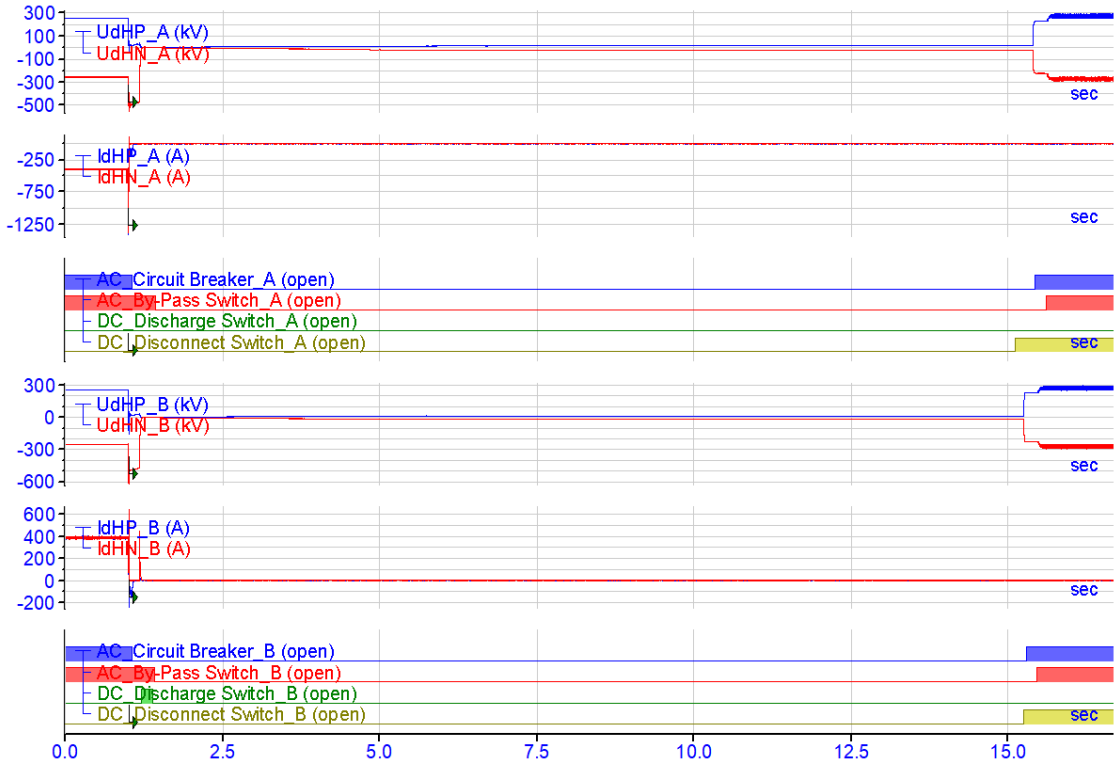


Figure 33: Commissioning record of the changeover to STATCOM operation after a third consecutive pole-to-ground fault, link previously at reduced direct voltage: the DC disconnect switches open and both stations deblock at about 15 s; for each station the pole voltages, pole currents and the switch positions. Reproduced from [24], Figure 9

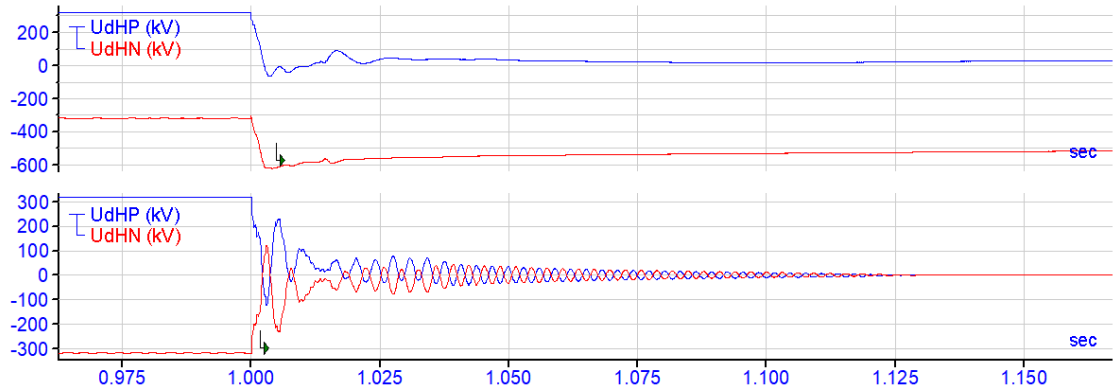


Figure 34: Pole voltages at the Thrissur cable termination within the first 150 ms of a fault on the overhead line near Pugalur: single pole-to-ground fault (upper) and pole-to-pole fault (lower). Reproduced from [24], Figure 6

6.11 Transfer to a single-link scheme

The recovery times above hold for the converter type and switchgear described; the supplier states that a substantial reduction requires a different converter topology that makes the mechanical switches unnecessary [24]. The power compensation during a first restart is specific to the two-link project: a scheme with one link has no parallel path, and the loss of transmission until deblocking — 0.86 to 1.25 s in the records of Section 6.6 — and the ramp to pre-fault power within 2 s act on the AC network at both ends. For such a project the supplier names other measures to limit that impact as a requirement [24].

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A Two-Port Networks

This appendix is background material. Section 5 is self-contained and does not depend on it; it records the origin of that section's equations and the conditions under which they hold.

The chain of matrices used in Section 5 is not a power-transmission construction. It is the standard description of any linear network reached through two terminal pairs, and it was in use for telephone repeaters and filters before it was applied to power lines. Stated on its own terms, it separates the results that are properties of a transmission line from those that hold for any two-port.

The description has two properties that account for its use. Two ports and four complex numbers describe the terminal behaviour completely, whatever the internal arrangement; and networks in cascade multiply, so a chain of elements of any length reduces to one matrix of the same size.

A.1 The port concept

A *port* is a pair of terminals through which the current entering one terminal equals the current leaving the other. That equality is the port condition, and it is an assumption about how the network is connected, not a property of the network itself: it fails if a third path — a stray capacitance to earth, a shared return, a ground loop — carries current between the two sides. A two-port is a network with exactly two such pairs, Figure A.1.

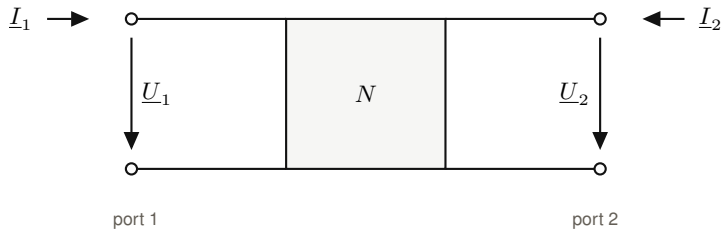


Figure A.1: A two-port and its terminal quantities, drawn in the convention in which both port currents enter the network

Four quantities appear at the terminals: $\underline{U}_1$, $\underline{I}_1$, $\underline{U}_2$ and $\underline{I}_2$. The network imposes two linear relations between them, so any two may be chosen as independent and the other two follow. Which two are chosen is the only difference between the parameter sets below.

A.2 Parameter sets

Table A.1 lists the six choices. The impedance, admittance and the two hybrid sets are drawn in the convention of Figure A.1, with both currents entering the network. The chain sets reverse the reference direction at port 2, so that $\underline{I}_2$ leaves the network towards whatever is connected next — without that reversal, cascading would alternate in sign at every junction. Comparing results from two sources that differ only in this convention is the most common error in using them.

The sets are equivalent wherever they all exist, and each is undefined for some networks: an ideal transformer has no impedance matrix, an ideal series element no admittance matrix. The chain matrix exists unless the network cannot transfer anything between the ports at all.

A.3 Chain parameters and cascading

The chain form writes port 1 in terms of port 2,

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \underline{T} \cdot \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix}, \qquad \underline{T} = \begin{bmatrix} \underline{A} & \underline{B} \\ \underline{C} & \underline{D} \end{bmatrix} \tag{A.1}$$

with $\underline{I}_2$ leaving the network. The four entries have a meaning one at a time. With port 2 open, $\underline{I}_2 = 0$ and $\underline{A} = \underline{U}_1/\underline{U}_2$ is the reciprocal voltage ratio, while $\underline{C} = \underline{I}_1/\underline{U}_2$ is a transfer admittance. With port 2 shorted,

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Set	Relation	Typical use
Impedance $\underline{Z}$	$\underline{U} = \underline{Z} \cdot \underline{I}$	Networks joined in series; open-circuit measurements
Admittance $\underline{Y}$	$\underline{I} = \underline{Y} \cdot \underline{U}$	Networks joined in parallel; nodal analysis
Hybrid $\underline{h}$	$(\underline{U}_1, \underline{I}_2)$ from $(\underline{I}_1, \underline{U}_2)$	Amplifying devices, where the input is a current and the output a voltage
Inverse hybrid $\underline{g}$	$(\underline{I}_1, \underline{U}_2)$ from $(\underline{U}_1, \underline{I}_2)$	The dual of the above
Chain $\underline{T}$ (ABCD)	port 1 from port 2	Elements in cascade; transmission paths
Inverse chain	port 2 from port 1	The same, traversed in the other direction

Table A.1: The six parameter sets of a two-port, distinguished only by which two terminal quantities are taken as independent [19], [20]

$\underline{U}_2 = 0$ and $\underline{B} = \underline{U}_1/\underline{I}_2$ is a transfer impedance, $\underline{D} = \underline{I}_1/\underline{I}_2$ the reciprocal current ratio. So $\underline{A}$ and $\underline{D}$ are dimensionless, $\underline{B}$ is an impedance and $\underline{C}$ an admittance — which is why the four cannot be compared with each other as magnitudes.

The reason this set is the one used for transmission is cascading. If two two-ports are connected so that port 2 of the first is port 1 of the second, Figure A.2, the intermediate quantities are shared and eliminate, leaving

$$\underline{T} = \underline{T}_a \cdot \underline{T}_b, \qquad \text{and in general} \qquad \underline{T} = \prod_{i=1}^{n} \underline{T}_i \tag{A.2}$$

in the order the elements are traversed. The product does not commute: reversing two elements of a chain generally changes the result, which is the algebraic statement that it matters where in a path an element sits. A compensating element at the middle of a line and the same element at its end are different networks, and (A.2) is what distinguishes them.

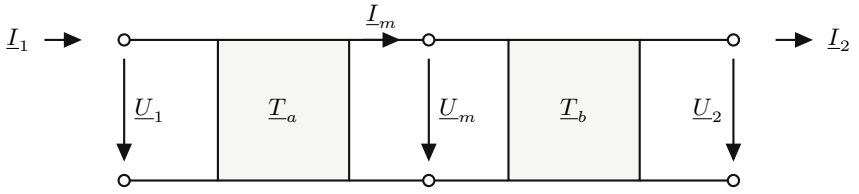


Figure A.2: Two two-ports in cascade: the intermediate quantities are common to both and eliminate, so the chain matrices multiply

A.4 Other interconnections

Cascade is one of five ways two two-ports can be joined, and each has its own parameter set in which the combination is a simple sum, Table A.2. The pattern is the same throughout: pick the set whose independent variables are the ones the connection forces to be shared, and the matrices add.

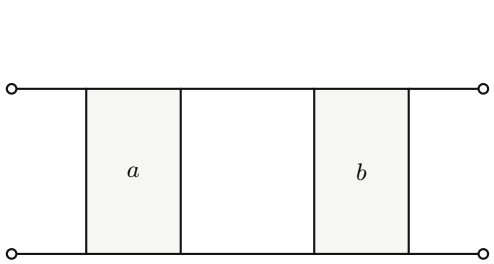
The cascade differs from the other four, and the word “series” is applied to both arrangements in ordinary use. In a cascade the networks are passed through one after another and the matrices *multiply*; in a series connection they sit side by side with their ports stacked, both carry the same currents, and the matrices *add*, Figure A.3. A capacitor bank inserted in a line is a cascade, not a series connection of two-ports, even though the capacitor is in series with the conductor: what is in series there is a circuit element within one port, not two ports with each other.

The addition rules carry a condition that the cascade does not. Stacking two networks can destroy the port condition inside them: current that entered one terminal of a sub-network may return through the other sub-network rather than through its partner terminal, and then the sub-network no longer has the matrix it was given alone, so the sum is wrong. Whether a particular pairing is safe is settled by a test due to Brune, applied to the two networks as connected [20]. Two cases are safe without testing: either sub-network having an isolating transformer at one port, and — in power work the common case — two networks that

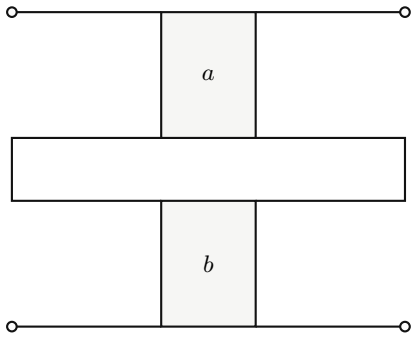
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Connection	Shared	Rule	Where it is met
Cascade	port 2 of one is port 1 of the next	$\underline{T} = \underline{T}_a \cdot \underline{T}_b$	Line sections, a series bank between two half-lines, a transformer ahead of a line
Series	both currents	$\underline{Z} = \underline{Z}_a + \underline{Z}_b$	Windings in series, impedances in a common current path
Parallel	both voltages	$\underline{Y} = \underline{Y}_a + \underline{Y}_b$	Parallel circuits of a double-circuit line, transformers in parallel
Series-parallel	$\underline{I}_1$ and $\underline{U}_2$	$\underline{h} = \underline{h}_a + \underline{h}_b$	Feedback arrangements in amplifiers
Parallel-series	$\underline{U}_1$ and $\underline{I}_2$	$\underline{g} = \underline{g}_a + \underline{g}_b$	The dual of the above

Table A.2: The five interconnections of two two-ports and the parameter set in which each is additive [19], [20]



(a) cascade: $\underline{T} = \underline{T}_a \cdot \underline{T}_b$



(b) series: $\underline{Z} = \underline{Z}_a + \underline{Z}_b$

Figure A.3: Cascade and series connection of two two-ports: in the cascade the networks are traversed in sequence, in the series connection their ports are stacked

share a common return and are each symmetric about it. The cascade needs no such test, because it adds no new connection between terminals that were not already a port.

The parallel rule is the one met most often in transmission. Two identical circuits on the same route, each with parameters $\underline{A} \ldots \underline{D}$, give

$$\underline{A}_{\parallel} = \underline{A}, \qquad \underline{B}_{\parallel} = \frac{\underline{B}}{2}, \qquad \underline{C}_{\parallel} = 2 \cdot \underline{C}, \qquad \underline{D}_{\parallel} = \underline{D} \tag{A.3}$$

which is the statement that two circuits halve the transfer impedance and so double the maximum transfer, while the voltage ratio is unchanged. Losing one of them restores $\underline{B}$, which is the contingency case of Section 5.7.3. Equation (A.3) follows from the admittance rule, and is quicker to use than converting twice.

A.5 Elementary two-ports

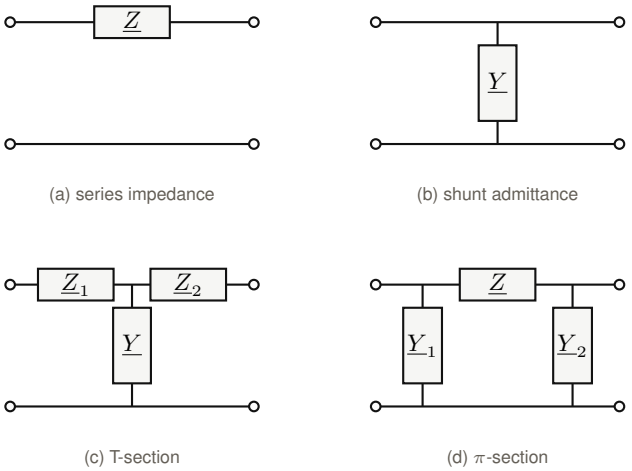
Every ladder network is built from two elements: an impedance in series with one rail and an admittance across the rails, Figure A.4(a) and (b). Their chain matrices are

$$\underline{T}_{series} = \begin{bmatrix} 1 & \underline{Z} \\ 0 & 1 \end{bmatrix}, \qquad \underline{T}_{shunt} = \begin{bmatrix} 1 & 0 \\ \underline{Y} & 1 \end{bmatrix} \tag{A.4}$$

and everything else in Table A.3 is a product of these two, taken in the order the elements are met.

The last row is the one that matters for a transmission line and is derived in Section 5.1.5; it is included here to show that it is the same object as the others, not a separate theory.

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(a) series impedance (b) shunt admittance

(c) T-section (d) π-section

Figure A.4: The two elementary two-ports and the two ladder sections built from them

Network	$\underline{A} \dots \underline{D}$	**Remark**
Series $\underline{Z}$	$1,\ \underline{Z},\ 0,\ 1$	$\underline{A} = 1$: no shunt path, so the open-circuit voltage ratio is unity
Shunt $\underline{Y}$	$1,\ 0,\ \underline{Y},\ 1$	$\underline{B} = 0$: the ports are directly connected
Symmetric π	$1 + \frac{\underline{Z}\cdot\underline{Y}}{2},\ \underline{Z},\ \underline{Y}\cdot\left(1 + \frac{\underline{Z}\cdot\underline{Y}}{4}\right),\ 1 + \frac{\underline{Z}\cdot\underline{Y}}{2}$	$\underline{Y}/2$ at each end; the nominal line section of (15)
Ideal transformer	$n,\ 0,\ 0,\ 1/n$	$\underline{B} = \underline{C} = 0$: a pure scaling of both quantities
Uniform line	$\cosh\underline{\theta},\ \underline{Z}_w \cdot \sinh\underline{\theta},\ \sinh\underline{\theta}/\underline{Z}_w,\ \cosh\underline{\theta}$	$\underline{\theta} = \underline{\gamma} \cdot l$; the limit of infinitely many π-sections

Table A.3: Chain matrices of the elementary two-ports and of the uniform line [19], [21]

A.6 Reciprocity, symmetry and losslessness

Three structural properties of a network show themselves directly in the four entries.

Reciprocity. A network built only from impedances, whatever their arrangement, satisfies

$$\underline{A} \cdot \underline{D} - \underline{B} \cdot \underline{C} = 1 \tag{A.5}$$

Physically: a source at port 1 and an ammeter at port 2 read the same as the source and the ammeter exchanged. It fails for networks containing an independent source or a non-reciprocal element such as a gyrator or a magnetised ferrite. Because the determinant is unity, the product in (A.2) preserves it, so any cascade of reciprocal networks is reciprocal.

Symmetry. If the network looks the same from both ports then

$$\underline{A} = \underline{D} \tag{A.6}$$

Every uniform line is symmetric; a π-section with unequal shunt branches is not, and neither is a bank placed anywhere other than at the midpoint. Symmetry is the condition under which the image impedance of Section A.7 is the same at both ends.

Losslessness. A network of pure reactances has $\underline{A}$ and $\underline{D}$ real and $\underline{B}$ and $\underline{C}$ purely imaginary. The departure of $\underline{A}$ from a real number is therefore a direct measure of the loss in a line, and is why the lossless approximations of Section 5 can be stated by dropping real parts rather than by rebuilding the model.

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A.7 Terminating a two-port

A two-port on its own is not a circuit. Closing port 2 with an impedance $\underline{Z}_L$ makes $\underline{U}_2 = \underline{Z}_L \cdot \underline{I}_2$, and (A.1) gives the impedance the source at port 1 sees:

$$\underline{Z}_{in} = \frac{\underline{U}_1}{\underline{I}_1} = \frac{\underline{A} \cdot \underline{Z}_L + \underline{B}}{\underline{C} \cdot \underline{Z}_L + \underline{D}} \tag{A.7}$$

Every statement about the effect of a network on the source behind it is a case of (A.7). With port 2 open it gives $\underline{A}/\underline{C}$, with port 2 shorted $\underline{B}/\underline{D}$.

One termination is distinguished. On a symmetric network there is a value of $\underline{Z}_L$ for which $\underline{Z}_{in} = \underline{Z}_L$. The network then presents its own termination impedance to its surroundings and may be repeated any number of times without changing them. Setting $\underline{Z}_{in} = \underline{Z}_L$ in (A.7) with $\underline{A} = \underline{D}$ gives the *image* or *characteristic* impedance

$$\underline{Z}_c = \sqrt{\frac{\underline{B}}{\underline{C}}}, \qquad \underline{\theta} = \operatorname{arcosh} \underline{A} \tag{A.8}$$

with $\underline{\theta}$ the image transfer constant, the complex number by which a quantity is scaled and shifted in passing through the network. Substituting the uniform-line row of Table A.3 returns $\underline{Z}_c = \underline{Z}_w$ and $\underline{\theta} = \underline{\gamma} \cdot l$: the wave impedance of a line and its electrical length are not line-specific quantities but this general pair, evaluated for that network. The natural power of Section 5.5 is the same statement in power terms — a line closed in its own image impedance.

A.8 Analogues in other domains

Nothing above used the fact that the quantities were electrical. The derivation needs only a pair of variables at each port whose product is power: one that is common to the elements in parallel and one that is common to the elements in series. Table A.4 lists the pairs for which the identical algebra is standard practice.

Domain	Across	Through	Where the chain form is used
Electrical	Voltage	Current	Lines, filters, matching networks, machine equivalents
Translational	Velocity	Force	Vibration isolators, drive trains, transducer chains
Rotational	Angular velocity	Torque	Shaft lines, gearboxes, torsional models of a turbine set
Acoustic	Sound pressure	Volume velocity	Ducts, horns, mufflers, loudspeaker enclosures [22]
Thermal	Temperature difference	Heat flow	Layered walls, heat sinks, cable thermal ratings

Table A.4: Domains in which the two-port algebra is used unchanged, with the pair of port variables whose product is power [20], [22]

A transducer is a two-port whose ports lie in two different rows of that table: a loudspeaker is an electrical port and an acoustic one, a motor an electrical port and a rotational one, a piezoelectric stack an electrical port and a translational one. Cascading them is the same matrix product, and it is how a complete electromechanical chain is reduced to one matrix.

Two further uses reach the same matrix from a different starting point.

Ray and beam optics. A sequence of lenses, mirrors and free spaces is handled by a 2×2 *ray transfer matrix* acting on the ray height and its slope, multiplied in the order the light meets the elements [23]. A thin lens is $[1, 0; -1/f, 1]$ — the shunt element of (A.4) with $\underline{Y} = -1/f$ — and a free space of length d is $[1, d; 0, 1]$, the series element. The resonator condition that the beam reproduce itself after one round trip is the image-impedance condition (A.8) in another vocabulary.

High-frequency networks. Above a few hundred megahertz a terminal voltage is not measurable and the scattering parameters are used in place of the chain matrix. They are a change of variables, not a different theory: the scattering matrix is the same network expressed in incident and reflected waves against a reference impedance, and converts back to $\underline{A} \dots \underline{D}$ in closed form. Cascading in that vocabulary uses the transfer-scattering matrix, which multiplies exactly as (A.2) does.

A.9 Assumptions

The description rests on five assumptions. All five hold throughout this report and are therefore not restated where it applies them.

- **Linear and time-invariant.** The relations are linear with constant coefficients. A saturating transformer core, a switching converter or a thyristor-controlled bank is not described by one matrix; it is described by one matrix per operating point, or by a model outside this framework.
- **One frequency.** The entries are complex numbers at a stated frequency. A chain evaluated at 60 Hz says nothing about its behaviour at 20 Hz, which is why the sub-synchronous screening of Section 5.8.6 re-evaluates the same network over a range of frequencies rather than scaling one result.
- **The port condition holds.** Current entering a terminal leaves its partner. Earth-return paths and inter-circuit coupling break it, and a line that carries zero-sequence current is a multi-port, not a two-port.
- **No internal sources.** Independent sources inside the network add a constant term that no 2×2 matrix carries. They are handled by superposition, or by moving them outside the two-port.
- **Balanced three-phase, for the power case.** The single-line two-port of Section 5 is the positive-sequence network. Unbalance requires the sequence networks separately.

Within those bounds the description is exact. The equations of Section 5 are not an approximation to a line but the general two-port result with the line's own $\underline{A} \dots \underline{D}$ substituted.

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